Horimetrik

Numbers with a Horizon

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with mathematical collaboration by Claude (Anthropic)
and ideas contributed by GPT (OpenAI)

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Translated from the German original (`buch/` in the project repository).
Where the two versions differ, the German text is authoritative.

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Preface: What happens if you flip the exponents?

This book was not born at a desk. It began in a child's bedroom, in the dark, while I was putting my daughter to sleep. I lay next to her, determined not to doze off myself – only to keep her company – and nothing keeps a mind awake as reliably as a thought that won't let go. For me such thoughts are numbers; my head computes on its own. That evening I was practising mental arithmetic, out of a slightly stubborn resolve not to unlearn calculation in the age of computers: 728 divided by 4. The 700 is 7 times 100, hence 7 times 4 times 25 – divided by four that leaves 7 times 25, which is 175; the 28 trivially gives 7; together, 182.

Nothing about this is remarkable, except for what caught my attention while taking the number apart: how naturally I sort hundreds, tens, and units into separate boxes. From there it was a short walk through everything I knew about positional systems – binary numbers, the base-twelve system you can count on the segments of your fingers – to a question I had never seen asked anywhere: what happens if the counting bases change *inside* a single number? That such systems already existed – the factorial numbers – I did not know that night; the first literature search turned them up later. But ascending bases struck me as boring. I wanted it the other way round: the most possibilities at the far right.[1] Whoever takes that thought seriously runs straight into a necessity: such a system cannot count to infinity – infinity would have been nonsense here, so: finite. What at first looks like a defect is the birth of this book's central notion: every number lives inside a *horizon*, a finite space with a fixed number of places. My daughter was long asleep by then. My wife told me later I had been upstairs for a good hour; it had felt like twenty-five minutes.

That leading zeros change the numbers in such a system – the oddity that drives everything that follows – I did not yet know that night. It came to light the way ideas get tested nowadays: I sat down at the computer and explained the thought to GPT, asking it to check whether all of this already existed. The first verdict was dry: nonsense. But we kept at it, and after a while we realized together that it was not complete nonsense after all. In those conversations the zeros surfaced – zeros that suddenly mean something at the front of a number – along with many of the ideas this book picks up. One sentence from that time has stayed with me: 4 times 5 can be an area – and if the 4 is a scalar, a length; in both cases 20. It has only a marginal bearing on Horimetrik, but it made an impression – and it finds a late echo in Chapter 4. The name, by the way, was coined entirely without ceremony: "horizon arithmetic" (*Horizont-Arithmetik*) was too long for me. So: Horimetrik.

At some point the subject had outgrown casual conversation, and GPT and I agreed that it deserved an agent of its own. That is how Fable entered the race – the Claude model by Anthropic that appears on the title page as mathematical collaborator. Let me be honest: mathematically, I

[1] Strictly speaking no exponent gets flipped; what is reversed is the direction in which the bases grow – the weight ordering of the positional system. The title follows the original thought, not the terminology.

was left behind fast. My computer science degree taught me mathematics, minor subject included – but that was more than a decade ago. I understand the outlines, and the outlines were exciting enough: not because there was any application, but because it all seemed novel enough to be worth pursuing. Many fields of mathematics began without an application. Fable can do mathematics; so I had everything recomputed and a clean theory built, and over time more and more bridges to established mathematics appeared. GPT kept contributing ideas along the way. My thanks go to both – read for yourselves what the AI and I have found.

The book pursues two goals. First, it wants to develop the mathematics of this space honestly – with definitions, proofs, and also with the wrong turns we took along the way. Second, it wants to stay readable: every technical passage returns to the surface before the next one begins. Whoever reads only the formula-free sections should still understand what is going on.

Whether Horimetrik becomes a theory in its own right or remains a particularly instructive plaything was something we ourselves did not know while writing. That is precisely what this book is about.

Chapter 1

The Idea: Numbers That Know How Much Room They Have

1.1 An ordering nobody questions

Every positional number system rests on a silent convention. In the decimal system the last digit counts ones, the one before it tens, then hundreds – each position weighs ten times its neighbour. The binary system does the same in steps of two. And there is a less familiar but venerable system in which the bases are not constant but grow: the factorial system. There the last digit may only be 0 or 1, the next one 0 to 2, the one after that 0 to 3, and so on. The further left a position sits, the more it may count *and* the more it is worth.

In all these systems the same direction points upward: the position with the most freedom carries the most weight. This seems so self-evident that nobody ever says it out loud.

Horimetrik begins with the question of what happens when you reverse exactly that. What if, of all positions, the *tightest* one – the one that knows only 0 and 1 – gets the greatest weight? What if the positions grow ever more generous towards the right, while their influence keeps shrinking?

1.2 Enforced finiteness

The first consequence arrives immediately, and it is not a design decision but a necessity. An ordinary number system grows leftward into infinity: when a number gets too large, you prepend another position with an even greater weight – there is always room. In the reversed system this is impossible. On the left already sits the tightest position, the one with base two, and below two there is nothing. The system is *sealed* on the left.

Whoever works with a fixed number of positions therefore gets a finite supply of numbers. With one position there are two, with two positions six, with three twenty-four, with four one hundred and twenty – every additional position multiplies the supply, and the sizes that arise are precisely the factorials. Once the supply is exhausted, no carry will help: you need a *new system* with more positions, in which every old digit suddenly carries a different weight.

This finite living space of a number is what we call its *horizon*. What at first looks like a defect of the reversed system – you cannot simply keep counting – is in truth its most interesting feature: every number here carries not only a value but also the information of which space that value lives in. The totality of all horizons remains infinite; only each single one is finite. Infinity still exists – but as a staircase, not as a straight line.

1.3 The second find: the zeros

The leading zeros came later, and they came as a surprise. In every familiar system a prepended zero is pure cosmetics: 012 is 12. In the reversed system you prepend a zero – and thereby change the number.

An example with small numbers. In the horizon with two positions the digit string 12 means the value five. Prepend a zero and it reads 012 – but now in a horizon with three positions, in which all the weights are distributed differently. The value is six. One more zero: 0012, value seven. The digits 1 and 2 have not moved, and yet the number counts one further with every extension.

This is the moment at which a curiosity becomes an object of study. For if a leading zero changes the value, then there are suddenly *two different answers* to the most harmless of all questions: what does it mean to give a number more room?

- You can hold its *value* fixed – then its digits must reorganize themselves when moving into the larger horizon.
- You can hold its *shape* fixed – the digits stay put, a zero is prepended – then its value shifts.

In the decimal system these are the same operation. Here they are two, and they lead into different worlds. Almost everything this book studies lives in the gap between these two notions: the two identities of a Hori number that are visible at first (a third will reveal itself later), the ways of tidying up, the worlds of arithmetic, and the tensions between them.

1.4 What was actually invented here – and what was not

Honesty belongs here from the start: the building blocks of this system are all known mathematics. Number systems with mixed bases have existed for centuries, the factorial system is well studied, and combinatorics has long known that its spaces have factorial size. What is new – at least we have found it nowhere in this form – is the combination: the reversed reading direction inside a finite space, the horizon as part of the number, the meaning-carrying zeros, and the two competing ways to grow. Whether this combination can carry a theory of its own was the bet on which this project began. The following chapters lay the cards on the table.

Back to the surface.

If only one sentence of this chapter is to remain, let it be this: Horimetrik did not arise from a wish to write numbers differently, but from the question of what an ordering enforces once you reverse it. The answer was: finiteness – and from finiteness everything else followed on its own.

Chapter 2

Finite Horizons and Factorial Shells

2.1 An afternoon in a world of 24 numbers

Before we define, we calculate. We take three places. The first may be 0 or 1, the second 0, 1, or 2, the third 0 to 3. The weights run from the right: the last place counts units, the middle one counts fours (because the last place becomes "full" with its four possible values before the middle one advances), the first counts twelves. Whoever writes down all the permitted combinations in order counts from 0 to 23:

$$\begin{array}{cccc} 000 = 0 & 001 = 1 & 002 = 2 & 003 = 3 \\ 010 = 4 & 011 = 5 & 012 = 6 & 013 = 7 \\ 020 = 8 & 021 = 9 & 022 = 10 & 023 = 11 \\ 100 = 12 & 101 = 13 & 102 = 14 & 103 = 15 \\ 110 = 16 & 111 = 17 & 112 = 18 & 113 = 19 \\ 120 = 20 & 121 = 21 & 122 = 22 & 123 = 23 \end{array}$$

No gap, no duplicate: exactly $2 \cdot 3 \cdot 4 = 24$ numbers. At 123 everything stops – every place stands at its maximum, and building on to the left is impossible, because below base two there is nothing. We call this world of 24 places the horizon H_3. Whoever wants to reach 24 needs a new world: H_4 with $2 \cdot 3 \cdot 4 \cdot 5 = 120$ places, in which all the weights are different.

2.2 The definitions

Definition 2.1 (Horizon; Register D1)**.** For $n \in \mathbb{N}_0$ let $H_n = \{(a_1, \dots, a_n) \mid 0 \leq a_i \leq i\}$. Place i has base $i + 1$; hence $|H_n| = 2 \cdot 3 \cdots (n{+}1) = (n + 1)!$.

Definition 2.2 (Value; Register D2)**.** $V_n(a) = \sum_{i=1}^{n} a_i \, \dfrac{(n+1)!}{(i+1)!}$, equivalently recursively $v_0 = 0$ and $v_i = (i{+}1)v_{i-1} + a_i$ from left to right, with $V_n(a) = v_n$.

Theorem 2.3 (Register S1)**.** $V_n \colon H_n \to \{0, \dots, (n + 1)! - 1\}$ *is bijective.*

Proof. Induction on n. For $n = 0$ there is nothing to show. The recursion yields

$$V_n(a) = (n{+}1)\, V_{n-1}(a_1, \dots, a_{n-1}) + a_n, \qquad 0 \leq a_n \leq n;$$

division with remainder by $n + 1$ therefore determines a_n and the remaining digits uniquely (injectivity). The largest value is $(n{+}1)(n! - 1) + n = (n{+}1)! - 1$, and since domain and codomain both have $(n + 1)!$ elements, surjectivity follows. $\square$

The inverse map E_n – decomposing a value into its digits – is repeated division with remainder by $n{+}1, n, \dots, 2$, that is, exactly the algorithm by which one breaks seconds down into days, hours, and minutes, only with regularly growing rather than arbitrarily mixed bases.

2.3 Factorial shells

Every natural number x has a smallest home: the *minimal horizon* $\mu(x) = \min\{n : x < (n+1)!\}$. The natural numbers thereby split into *shells* $S_n = \{x : \mu(x) = n\}$ – and these are conceptually different from the horizons: the horizon H_n is the whole space and houses the shells S_0 to S_n; the shell S_n is its *new territory*, the values that only become possible with it.

Shell	Range	New values
S_0	only the 0	1
S_1	only the 1	1
S_2	2 to 5	4
S_3	6 to 23	18
S_4	24 to 119	96
S_5	120 to 719	600
S_6	720 to 5039	4320

The shell boundaries are number-theoretically striking places: for $n \geq 1$ the lower boundary of the shell S_n is the factorial $n!$ – divisible by *every* number from 1 to n. The number directly before it, $n! - 1$, is divisible by *no* number from 2 to n. The step from 719 to 720 thus switches, within a single successor, from maximal coprimality to maximal divisibility – and at precisely this point Horimetrik forces the move into a new world.

Example 2.4. The number 7 lives minimally in H_3 (since $6 \leq 7 < 24$) and is written there as 013. In H_4 it is written 0012, in H_5 as 00011, in H_6 as 000010 – and from H_7 onwards stably as $0 \cdots 07$: as soon as the horizon is large enough, the whole number fits into the last digit, and further leading zeros are at last genuinely harmless. We tell this life story at leisure in an interlude of its own later on.

With that the foundation is laid: finite worlds of factorial size, a bijection between digits and values, and shells with striking boundaries. What is still missing is the true protagonist – the second identity of every number. It steps onto the stage in the next chapter.

Interlude: What a Horizon Really Is

Back to the surface.

Let us pause for a moment, entirely without formulas.

A horizon is not a length and not a format. It is the space a number currently lives in – with a fixed, finite number of places that come from the factorials: two, six, twenty-four, one hundred and twenty. An ordinary number knows only its value. A Hori number knows its value and its space, and both belong to it the way a book has both its text and the shelf it stands on.

That sounds like a small thing, but it is a change of perspective. Our usual mathematics takes the infinite number line as given and places all numbers on it at once. Horimetrik builds staircases instead: each step is finite and completely surveyable, and the infinity lies not in any single step but in the promise that there is always a next one. Nothing about classical infinity is refuted by this – it is merely told differently: as a sequence of spaces rather than as one single space.

The rest of this book investigates what this way of telling costs and what it earns. It costs convenience: with every move from step to step, all digits change their meaning, and you must decide what the move is to preserve – the value or the shape. It earns structure: out of exactly this decision arise the two identities, the two worlds of arithmetic, and the theorem that you cannot live in both at once.

Chapter 3

Two Ways to Grow: Shape or Value

3.1 The moment of the fork

Take the seven in its home H_4: there it is written 0012. Now we give it one more place. What is that supposed to mean?

The first answer is the naive one: we put a zero in front.

$$0012 \longmapsto 00012.$$

The digits stay put – but in H_5 every place except the last carries a different weight, and the value slips from 7 to 8. The *shape* was preserved, the value was not.

The second answer is the cautious one: we want the number to stay the same, and we re-encode the 7 in H_5:

$$0012 \longmapsto 00011.$$

The *value* was preserved, the shape was not.

In the decimal system both answers are the same operation – which is why we never noticed that they are two questions. In Hori space they part ways, and we give them names: the structure-preserving – equivalently we say: shape-preserving – extension Z puts the zero in front, the value-preserving lift L re-encodes.

Definition 3.1 (Register D5)**.** $Z_n(a_1, \dots, a_n) = (0, a_1, \dots, a_n)$ and $L_n(a) = E_{n+1}(V_n(a))$, where E_{n+1} is the encoding into H_{n+1}. We have $V_{n+1}(L_n(a)) = V_n(a)$, whereas Z_n in general changes the value.

3.2 Two staircases, two skies

Both operations join the finite horizons into an infinite staircase – but they are *different* staircases, and they lead to different places. The question in each case is: which elements of consecutive horizons count as "the same"?

Under L two representations count as the same if they have the same value: 013 in H_3, 0012 in H_4, 00011 in H_5 are a single object – the seven. Under Z they count as the same if they have the same shape: 12, 012, 0012 are a single object – the pattern "one-two", whose value wanders from step to step.

Theorem 3.2 (Register S8)**.** *The limit space of the staircase under L is $\mathbb{N}_0$, the set of ordinary natural numbers. The limit space under Z is the space of all shapes – formally: the semiring of* *polynomials with non-negative integer coefficients in the basis of falling factorials, which the next chapter introduces.*

Proof idea. Under L precisely the representations of the same value are identified (the encoding is injective), and every value occurs – the classes *are* the natural numbers. Under Z precisely those digit strings are identified that differ by leading zeros, and every shape occurs from its smallest admissible horizon onwards – the classes are the shapes. The complete proof via directed systems is given in Appendix B. □

Example 3.3. The two staircases, side by side. Preserving value, the seven remains itself and changes shape; preserving shape, the pattern "one-two" remains and changes value:

World	H_2	H_3	H_4	H_5
value-preserving (always 7)	—	013	0012	00011
shape-preserving (always "12")	$12 = 5$	$012 = 6$	$0012 = 7$	$00012 = 8$

Note the crossing: 0012 appears in both rows – once as the seven's second home, once as the pattern's third life. The same element, two readings.

This is the first theorem in this book that one may read philosophically: *the same finite spaces generate different infinities, depending on which notion of consistency one chooses between the steps.* The familiar number line is one of these infinities – the value-preserving one. The other, the shape-preserving one, is mathematically just as legitimate; only, in the ordinary number system nobody has ever been able to tell it apart from the first, because there the two staircases coincide.

3.3 The third staircase

For a long time we thought there were exactly two ways to grow. Then an outside mathematician read a short description of this system – and inadvertently anchored the ladder of bases at the other end. Out of his "misunderstanding" fell a third staircase.

For one can also append a zero to a number *on the right.* All digits then keep their position counted from the left, and a strikingly clean rule holds: the value does not multiply out of control but scales *exactly* by the factor $n + 2$ – 12 in H_2 (value 5) becomes 120 in H_3 (value 20). What stays invariant is neither the value nor the shape but a third quantity: the *factorial fraction* $F = \sum_i a_i/(i + 1)!$, since we always have

$$V_n(a) = (n + 1)! \cdot F(a).$$

Every Hori number is thus also a fraction between zero and one, times the size of its world – and this fraction system is an old acquaintance: it is Cantor's factorial base for fractions, a good century and a half old.

With that, Hori space has three staircases, three invariants, three skies:

Extension	Invariant	Value	Limit space
value-preserving (L)	value	$\times 1$	natural numbers
shape-preserving (Z, left)	polynomial	uneven	structure space
fraction-preserving (R, right)	fraction F	$\times(n + 2)$	Cantor fractions

And the objection "but this has been known since Cantor" thereby receives a precise answer: it hits the third staircase completely – and the other two, together with everything this book proves about their interplay, not at all.

3.4 The eigenhorizon: where the fork heals

For every single number there is a point at which the difference between Z and L vanishes. If the entire value sits in the last digit – the number n as $0\cdots 0n$; we call this shape its *terminal representation*, and it becomes possible from the horizon H_n onwards –, then the zero extension is value-neutral: shape and value grow together from there on. We call H_n the *eigenhorizon* of n. Below it the number is rebuilt at every move; from there on it only grows quietly to the left. One can compute exactly when a leading zero is value-neutral: precisely when all digits except the last are zero – and the next chapter turns this into a one-liner using the horizon derivative.

What this feels like for a concrete number is told in the interlude "The life story of the number seven".

Chapter 4

Structure Polynomials and the Horizon Derivative

4.1 The shape acquires coordinates

So far the "shape" of a Hori number was nothing but its digit picture. Now we give it an algebraic form – and an intuition becomes a computational tool.

The key are the *falling factorials*

$$X^{\underline{0}} = 1, \qquad X^{\underline{d}} = X(X-1)\cdots(X-d+1),$$

the polynomials that count "ordered selection without replacement". Reading the digits of a Hori number from right to left as coefficients in this basis produces its *structure polynomial*:

$$a = (a_1, \ldots, a_n) \quad \longmapsto \quad P_a(X) = \sum_{d=0}^{n-1} a_{n-d}\, X^{\underline{d}}.$$

Theorem 4.1 (Evaluation theorem; Register S2)**.** $V_n(a) = P_a(n+1)$.

Proof. For $d = n-i$ we have $(n+1)^{\underline{d}} = (n+1)!/(i+1)!$ – exactly the weight of place i. Substituting yields the value formula. □

A small computation with a large consequence: the value of a Hori number is *the evaluation of its shape at the place of its horizon.* The seven in H_4, that is 0012, has the shape $P(X) = X + 2$; its value is $P(5) = 7$. Prepend a zero and the shape remains $X + 2$ – only the evaluation point moves: $P(6) = 8$. The mysterious value shift caused by leading zeros is no longer a riddle but simply this: *the same polynomial, evaluated one place further on.*

Two precautions. First: only nonnegative integer coefficients are admissible, and in horizon n only polynomials with $d+c_d \le n$ at all occupied places ($c_d > 0$) – that is the digit bound in polynomial language; the quantity $\sigma(P) = \max_{c_d>0}(d + c_d)$ – the maximum runs over the occupied places, and for the zero polynomial we agree that $\sigma(0) = 0$ – is called the structure horizon (Register D4). Second: the polynomial alone is *not* the whole Hori number. 12, 012 and 0012 all have the shape $X + 2$, yet they are three different elements – only the pair of horizon and polynomial carries everything.

4.2 The horizon derivative

How strongly does a leading zero change the value? The answer is the discrete derivative, known in analysis as the forward difference:

$$\Delta P(X) = P(X+1) - P(X), \qquad \Delta\, X^{\underline{d}} = d\, X^{\underline{d-1}}.$$

The second formula is the discrete counterpart of $(x^d)' = d\,x^{d-1}$ – the falling factorials are to the calculus of differences what the powers are to differentiation.

Theorem 4.2 (Register S7)**.** *The value change under a structure-preserving extension is* $\Delta P(n+1)$*. In particular, a leading zero is value-neutral exactly when* $\Delta P = 0$*, that is, when* P *is constant – these are precisely the numbers at or above their eigenhorizon.*

Example 4.3. The shape $X+2$ (digits $\ldots 12$) has derivative 1: its values climb by exactly one with each move, $5, 6, 7, 8, \ldots$ The shape $X^{\underline{2}}$ – that is the twelve in H_3, digits 100 – has derivative $2X$: its values jump $12, 20, 30, 42, \ldots$, with increments $8, 10, 12$ that grow themselves. A constant such as the shape 7 has derivative 0 and never stirs.

With this, the eigenhorizon from Chapter 3 is no longer an intuitive notion but an equation: the *shapes* with horizon derivative zero are exactly the constant polynomials – read as shapes, the natural numbers. (Strictly speaking: as elements, the constants form many horizontal representatives of the same value; only in the shape-preserving limit space of Chapter 3 does this become exactly one copy of $\mathbb{N}_0$.) Everything else in the Hori space has a genuine "velocity": a shape of degree 1 grows by a constant amount per horizon step, degree 2 accelerates, and so on. The degree of the polynomial is the *Hori dimension* of a shape (Register D7).

4.3 Length times length is area – now exact

Dimension behaves additively under multiplication: $\dim(PQ) = \dim P + \dim Q$ for $P, Q \neq 0$. What is a mnemonic at school – length times length gives area – is here a theorem about degrees. And one can compute with shapes directly. In H_3 the four has the shape X (digits 010), the five the shape $X+1$ (digits 011). Their shape product is

$$X(X+1) = X^{\underline{2}} + 2X$$

(since $X^2 = X^{\underline{2}} + X$), hence the digit string 120 – and indeed: 120 in H_3 has the value $1 \cdot 12 + 2 \cdot 4 + 0 = 20$. The multiplication $4 \cdot 5 = 20$ has happened here without a single carry, purely by combining the shapes. The product formula of the falling factorials carries a pretty side meaning along the way: $X \cdot X = X^{\underline{2}} + X$ splits all ordered pairs into those with distinct and those with equal entries.

4.4 What operations cost: the horizon calculus

Every shape needs a horizon large enough for it – the quantity that measures this is the structure horizon σ. With it, every operation acquires a *price*: how much capacity must the target space have so that the result is guaranteed to fit there? For an operator T we ask for the worst case

$$\Gamma_T(r, s) = \max\{\sigma(T(P, Q)) \mid \sigma(P) \leq r,\ \sigma(Q) \leq s\}$$

(Register D10; for unary operators, correspondingly $\Gamma_T(r)$). At first sight a monster: already for $r = s = 8$ more than a hundred billion pairs would have to be checked. One suffices.

Theorem 4.4 (Extremal principle; Register S39). *In every horizon r there is the maximally filled shape M_r – every digit at the limit, $M_r(X) = \sum_{d<r}(r-d)\, X^{\underline{d}}$; we also call it the maximal polynomial of the horizon. For every operator that answers growing coefficients with growing coefficients (all those composed from addition, multiplication and Δ do), we have*

$$\Gamma_T(r,s) = \sigma\big(T(M_r, M_s)\big).$$

The worst case among untold billions of candidates is always the same: all digits full. The proof (Appendix B) is one line of monotonicity – but the consequence is a *calculus*: addition costs exactly $r+s$; multiplication costs $\beta(r,s)$, the product height, factorially growing at its core, which we shall meet again in Chapter 6 as a counting sequence. And the derivative delivers the biggest surprise of the calculus:

Theorem 4.5 (Difference overflow; Register S40). *For $r \geq 2$,*

$$\max_{\sigma(P)\leq r} \sigma(\Delta P) = \left\lfloor \tfrac{(r+1)^2}{4} \right\rfloor - 1.$$

Pause for a moment: Δ *lowers* the degree – and still lets the capacity swell quadratically. Already at $r = 8$ a single difference lifts the structure horizon from 8 to 19. The witness is a shape with a single digit in the middle: the derivative multiplies each coefficient by its degree, and the most expensive compromise between "high degree" and "large coefficient" lies exactly in the middle – hence the term $\lfloor (r+1)^2/4 \rfloor$, the largest product of two numbers with a fixed sum. The lesson: the degree measures the *dimension* of a shape, σ measures its *capacity*, and the two can jump in opposite directions. An ordinary degree analysis is blind to this effect.

One operation is still missing from the calculus, the most delicate one: *substituting* one shape into another, $P \circ Q$. That this is allowed at all is itself a theorem – it is by no means clear from the outset that substitution again produces nothing but admissible, integral, nonnegative digits.

Theorem 4.6 (Composition closure; Register S41). *Substituting a shape into a shape yields a shape again. Substitution, too, obeys the extremal principle: $\Gamma_\circ(r,s) = \sigma(M_r \circ M_s)$.*

The proof (Appendix B) is the combinatorial centrepiece of this chapter: one reads Q as a stock of blueprints – each digit a_k provides a_k templates, to be mounted on k distinct slots –, then $P \circ Q$ counts tuples of such assemblies, and the required divisibility by $m!$ comes from the fact that relabelling the slots cannot fix any choice of assemblies. The cost of substitution, however, is of an entirely different calibre from everything so far: the diagonal $\Gamma_\circ(r,r)$ begins with

$$1, \quad 4, \quad 23, \quad 6541, \quad 477\,149\,751, \quad \ldots$$

– substitution multiplies the degrees *and* makes the coefficients explode; this is the ninth and by far the fastest-growing counting sequence of this project (Chapter 6). The most harmless special case, the mere shift $P(X) \mapsto P(X{+}1)$, costs exactly $r + \lfloor r^2/4 \rfloor$ – to the point exactly as much as one derivative one horizon higher (Register S42), a small duality the calculus throws in for free.

4.5 What the shapes count: the counting width

To close the chapter, the question one should always put to an algebraic object: does it count something? The answer arrived as an external classification of the project and is so far Horimetrik's best bridge into an established field – *counting complexity.*

The key is the basic meaning of the falling factorial: $n^{\underline{d}}$ counts the ordered d-tuples of *distinct* objects from a stock of n. If now n is the number of solutions of some search problem – in computer science one says: of its *witnesses* –, then a shape $P = \sum_d c_d X^{\underline{d}}$ reads as a blueprint for new counted objects: c_0 fixed special objects, c_1 colours for each individual solution, c_2 variants for each ordered pair of distinct solutions, and so on. $P(n)$ is exactly the number of these coloured witness tuples. This harmless observation has a precise consequence:

Theorem 4.7 (Hori #P theorem; Register S43)**.** *If f is the witness-counting function of an arbitrary NP problem and P a shape, then $P \circ f$ is again the witness-counting function of an NP problem. Every shape is thus a closure operator of the counting class #P.*

The proof (Appendix B) is exactly the blueprint reading: guess the degree, guess the colour, guess a tuple of distinct witnesses. Honesty first: the *qualitative* side of this statement is known – polynomials with nonnegative integer coefficients in the *binomial* basis are called *binomial-good* in the literature and are characterized as the (relativizing) #P closure operations. Because of $X^{\underline{d}} = d!\binom{X}{d}$ our shapes are a *proper subclass* of these: divisibility by $d!$ is exactly the difference between ordered tuples and unordered selections. What the literature apparently does *not* have is the quantitative layer – and Horimetrik supplies it for free:

Suddenly all the quantities of this chapter mean something. The *addition* of shapes is the disjoint union of two counting constructions. *Multiplication* couples two constructions, and the coefficients of its product formula count exactly the overlap patterns of two groups of witnesses – so the growth of β measures how the variety of overlaps explodes under coupling. The *horizon derivative* counts which new structures a single additional witness makes possible (d positions times remaining tuples: exactly $d\,n^{\underline{d-1}}$) – and the difference overflow says: a new witness lowers the tuple order, yet can inflate the variety of variants quadratically. And σ itself? Draw the coefficient board – column d with c_d boxes –, then $\sigma(P) = \max(d + c_d)$ is the smallest *staircase* into which the board fits: the maximum of “how many witnesses at once” and “how many variants of them”. We call this quantity the *counting width* of a construction. Every horizon r is thus a finite, completely enumerable family of exactly $(r+1)!$ #P closure operators, and the horizon calculus of Section 4.4 gives the optimal figures for how the counting width grows under uniting, coupling, extending and substituting.

Two sentences of demarcation, so that no false magic arises: in this neighbourhood lives a weighty open conjecture (even its simplest version would imply $\mathrm{P} \neq \mathrm{NP}$), and our operators lie expressly on the *well-understood* side of this landscape – Horimetrik solves nothing here. What it contributes is more modest, and perhaps usable for precisely that reason: turning an allowed-or-not-allowed, for the first time, into a *how wide*.

Back to the surface.

What this chapter has delivered, in one sentence: the shape of a Hori number is a polynomial, its value is that polynomial evaluated at the place of its horizon, and the puzzling effect of leading zeros is simply the derivative of this polynomial. The ordinary numbers are the shapes with derivative zero – the only inhabitants of the Hori space who do not care where they live. All the others move when their world grows. Every arithmetic operation has a price in capacity that can be stated exactly, and it is always the same extreme object that pays it: the shape filled to the brim. And at the end it turned out that the shapes count something – coloured groups of distinct solutions of a search problem – and that σ measures the width of these counting constructions. In the next chapter we let these moving objects do arithmetic – and run into the question of what should then happen to their past.

Interlude: The Life Story of the Number Seven

Back to the surface.

Let us accompany a single number through all of its homes.

The seven is born into the world with twenty-four places – earlier is impossible; the world before it ends at five. There it is called 013. *It is a cramped apartment: three digits, all of them carrying something, nothing is decoration.*

Then it moves, world after world, always value-preserving – it wants to remain the same seven, after all. And with every move it has to furnish itself anew: in the world with one hundred and twenty places it is called 0012, *in the next one* 00011, *then* 000010. *Lay the moves side by side and you see a restless pattern – the digits wander, swap, shrink. No two moves are alike, and nothing in the spelling betrays at first glance that it is always the same number.*

Until the world with forty thousand three hundred and twenty places. There something new happens: for the first time the seven fits entirely into the last digit – it is now called 0000007, *and you can see its value again. From this move on, nothing ever changes again. Each further world merely hangs one more silent zero in front:* 00000007, 000000007, *and so on forever. The seven has arrived; we say: it has reached its eigenhorizon.*

Every number has this biography: a restless youth, in which each larger space means a fresh negotiation of identity, and a quiet adulthood, from which point on more space is simply more space. Large numbers have a long youth – the number seven hundred and twenty, for instance, keeps being rebuilt until world number seven hundred and twenty, and along that long road it temporarily wears shapes nobody would credit it with: in the ninth world it is simply a single one in the fourth position from the right, because ten times nine times eight is seven hundred and twenty. Exactly this double nature – unwieldy by value, plain by shape – later turned out to be the key to one of the main theorems.

Only one thing needs to be remembered from all this: in the Hori space, "the same number" is not a matter of course but a decision. You can hold the value fixed and sacrifice the shape, or the other way round. How much that decision costs is no longer a matter of taste but a theorem – but that belongs in the next chapter.

Chapter 5

Arithmetic with Memory: the Semiring

Up to this point we have only looked at Hori numbers and shifted them about. Now we want to compute with them. That sounds harmless, but it is the point at which Horimetrik demands its first genuine decision: when two Hori numbers are added, which horizon does the result receive?

In the ordinary number system this question never arises – there a result has no horizon, only a value. Here, by contrast, the horizon is part of the number. A rule of computation must therefore consist of two parts: a rule for the value and a rule for the horizon. The horizon rule we call a *selector*.

5.1 The first failure and what it teaches

The most obvious selector is the conservative one: the result inherits the largest horizon involved, enlarged if necessary to what the result value minimally requires. For addition this works flawlessly. For multiplication it fails – and it fails at the most innocent of all numbers.

Dead end. If one multiplies conservatively, then in general $(a \otimes b) \otimes c \neq a \otimes (b \otimes c)$ as soon as a zero is involved. The zero erases the *value* of every product, but the conservative selector keeps dragging along the *horizons* of the intermediate steps. Depending on when the zero strikes, the result remembers different computational routes – same value, different horizons, no freedom of bracketing. The lesson: a horizon carries computational history, and computational history does not automatically get along with the laws of arithmetic.

Example 5.1. Concretely, with the conservative multiplication (horizon = maximum of those involved, enlarged if necessary): let $a = b = (2, 5)$ – the five with horizon 2 – and $c = (1, 0)$ the zero with horizon 1. Bracketed on the left: $a \otimes b = (4, 25)$, since 25 needs horizon 4; then $(4, 25) \otimes c = (4, 0)$. Bracketed on the right: $b \otimes c = (2, 0)$, then $a \otimes (2, 0) = (2, 0)$. Both routes end at the value zero – but once with horizon 4, once with horizon 2. $(4, 0) \neq (2, 0)$: no associativity.

5.2 The value-side semiring

The way out is a division of labour: addition may remember, multiplication must forget. (One convention in advance, valid for the whole book: we call an arithmetic *law-abiding* if both operations are associative and commutative, multiplication distributes over addition, and an additively neutral element exists. A multiplicative identity and absorption by the additive zero we do *not* require in general, but only where they are explicitly named – both are about to meet us as genuine distinguishing features.)

Theorem 5.2 (value-side semiring)**.** *On the Hori space* $\mathcal{H} = \{(n,x) \mid n \geq \mu(x)\}$ *let, in value coordinates* (n,x)*,*

$$(n,x) \oplus (m,y) = \big(\max(n,m,\mu(x+y)),\, x+y\big), \qquad (n,x) \otimes (m,y) = \big(\mu(xy),\, xy\big).$$

Then $\oplus$ *and* $\otimes$ *are commutative and associative,* $\otimes$ *distributes over* $\oplus$*, and* $(0,0)$ *is neutral for* $\oplus$ *and absorbing for* $\otimes$*.*

The proof is astonishingly short; its single core is the monotonicity of the minimal horizon: more value never needs less room (in full in Appendix B). What is remarkable is what this semiring *lacks*: an identity. There is no element that leaves everything unchanged under multiplication. Instead, for the Hori number one – that is, $(1,1)$, the one in its smallest home – we have the equation

$$h \otimes (1,1) = K_V(h) \qquad \text{with} \quad K_V(n,x) = (\mu(x),\, x).$$

Multiplication by one is the value compactification.[1] The one acts not as a neutral element but as a *projector*: it presses every Hori number down to its compact form. The compact elements form a small semiring of their own underneath, and it is exactly the ordinary $\mathbb{N}_0$. The familiar natural numbers thus sit as a "corner" in the Hori space, and the way there is not an extra operation but multiplication by one.

Example 5.3. Let us work through it once. $(3,7) \oplus (2,5)$: value 12, horizon $\max(3,2,\mu(12)) = \max(3,2,3) = 3$, hence $(3,12)$ – the largest horizon involved survives. $(3,7) \otimes (2,5)$: value 35, horizon $\mu(35) = 4$, hence $(4,35)$ – the past is forgotten. And the projector: the seven with an unnecessarily large horizon, $(5,7)$, times the one $(1,1)$ gives $(\mu(7),7) = (3,7)$ – multiplication by one has tidied up.

Back to the surface.

What has happened here? We have found an arithmetic in which numbers remember their past when added – the largest horizon involved survives – and are born anew when multiplied. And the ordinary arithmetic we have known since primary school is contained within it: as the part of the Hori space that carries no past.

5.3 Excess: memory acquires a coordinate

For a long time it looked as though this forgetting multiplication had no alternative. The natural conjecture was that the laws of arithmetic destroy every other horizon rule. It is false, and the refutation is more instructive than a proof would have been.

The horizon of every Hori number splits canonically into two parts:

$$n = \underbrace{\mu(x)}_{\text{obligation}} + \underbrace{\varepsilon}_{\text{excess}}.$$

The obligatory part is the minimum the value needs; the excess is memory carried along voluntarily. This decomposition is not a convention but is uniquely determined by the factorial shells – and it decouples the question of the selector completely: one may compute on the excess *independently*.

[1] Terminological warning: "compactification" is not meant topologically here. Formally these are idempotent retractions onto the respective minimal horizon – we keep the intuitive word and ask topologically trained readers for their indulgence.

Theorem 5.4 (product principle)**.** *Every commutative semiring structure on the excesses yields a law-abiding Hori arithmetic: values compute in the ordinary way, excesses compute in the chosen structure.*

The product principle thus supplies a whole family of new arithmetics. Together with the forgetting semiring already constructed in Theorem 5.2 – which, mind you, is *not* a product construction: its addition selector compares the horizons themselves, not merely the excesses – three particularly natural arithmetics emerge:

	memory under +	memory under ×	identity
forgetting (Thm. 5.2; not a product)	maximum (of horizons)	dies	none
preserving (tropical)	maximum	adds up	$(1, 1)$
amplifying (arithmetic)	adds up	multiplies	$(2, 1)$

Example 5.5. Let $a = (5, 7)$ – obligation $\mu(7) = 3$, excess 2 – and $b = (3, 5)$ – obligation 2, excess 1. The product has value 35 with obligation $\mu(35) = 4$. The three arithmetics differ only in the excess of the result: forgetting $(4, 35)$ (excess 0), preserving $(4{+}2{+}1, 35) = (7, 35)$ (excesses add up), amplifying $(4{+}2, 35) = (6, 35)$ (excesses multiply: $2 \cdot 1 = 2$).

The preserving arithmetic has a curiosity: its zero does not absorb. If one multiplies a Hori number by zero, the value is wiped out, but the memory survives. The amplifying arithmetic, by contrast, is a fully fledged commutative semiring with identity and absorbing zero – and its multiplication does not erase memory wholesale but reckons with it arithmetically. Admittedly it too is not entirely faithful to memory: an operand with excess zero absorbs the other's memory, since zero times anything remains zero – and exactly this is where its absorbing zero comes from.

Back to the surface.

The question "which horizon does the result receive?" has turned into something far more beautiful: "what is to happen to the past when computation takes place?" And Horimetrik's answer is: that is yours to choose – forget, preserve, or amplify – and each choice yields a world that is perfectly free of contradiction within itself.

5.4 The duality: the same one, two projectors

Up to this point everything computed on the value side. But Hori numbers carry, besides their value, their shape as well (and, since Chapter 3, the fraction – of which more in Section 5.6) – and the shape side has an arithmetic of its own. Instead of the values one can add and multiply the *structure polynomials*; that is digit-wise computation without carries, in which, for instance, the structures of 4 and 5 directly give rise to the structure of 20.

Theorem 5.6 (duality theorem)**.** *The Hori space carries a second, structure-side semiring: addition conservative, multiplication compact – but measured against the structure horizon σ instead of the value horizon μ. Computation takes place in structure coordinates (n, P); via the digit bijection from Chapter 2, (n, x) and the pair consisting of n and the shape of x denote the same element. In this semiring*

$$h \boxtimes (1, 1) = K_S(h) \qquad \textit{with} \quad K_S(n, P) = (\sigma(P), P),$$

and K_S is compatible with both operations. The corner of this semiring is the space of structure polynomials.

The punchline deserves a paragraph of its own. It is the *same* Hori number $(1, 1)$ – the one – that carries the projector K_V on the value side and the projector K_S on the structure side. Two multiplications, one element. And the oldest result of this project – that the two tidying operations do not commute; the next section makes this formal – turns out to be the non-commuting of these two multiplications with one and the same one.

5.5 Tidying up: the commutator and the depth law

Because a whole thematic strand of this book rests on them, we now place the two projectors side by side (Register D6):

$$K_V(n,x) = (\mu(x),\, x), \qquad K_S(n,P) = (\sigma(P),\, P).$$

K_V tidies up *value-compactly*: same value, smallest horizon – the shape is recoded in the process. K_S tidies up *structure-compactly*: same shape, smallest horizon – the value migrates downwards with the evaluation point. Both are idempotent, but they do not commute, and the deviation is measurable. We call

$$\delta(h) = \operatorname{hor}\big(K_S K_V(h)\big) - \operatorname{hor}\big(K_V K_S(h)\big)$$

the *compactification commutator*; its negative values are called *depth*.

Example 5.7. The house spirit of this book: $h = (6,6)$, the six in terminal representation. K_V first: $(6,6) \mapsto (3,6)$; there the six has the shape $X{+}2$ with $\sigma = 2$, hence $K_S K_V(h) = (2,5)$ – horizon 2. K_S first: the constant shape 6 is already structure-compact, $K_S(h) = h$, then $K_V(h) = (3,6)$ – horizon 3. Hence $\delta(h) = 2 - 3 = -1$: the order of tidying up decides where you end up – and as what.

Upwards, δ grows comparatively fast; the smallest witness for $+2$ is the 720 as shape $X^{\underline{3}}$ in horizon 9 (the "single one in the fourth place" from the last interlude). Downwards, by contrast, δ is factorially sluggish, and this sluggishness obeys an exact law. Call

$$g(m) = m - \min\{s \mid M_s(m{+}1) \geq m!\}$$

the *shell gap* of the shell S_m: how far below m the capacity of a shape may lie whose value nevertheless still reaches S_m (M_s is the maximally filled shape from Chapter 4).

Theorem 5.8 (depth law, lower side; Register S33)**.** *For every shell $m \geq 3$ we have*

$$\min\{\delta(h) \mid \mu(\operatorname{val}(h)) = m\} = -g(m),$$

and an always serviceable witness – in general not the only one – is the terminal representation of the maximal-polynomial value $M_{m-g(m)}(m{+}1)$. The jump positions $m_g = \min\{m \mid g(m) \geq g\}$ form a counting sequence with proven asymptotics $m_g/(g{+}2)! \to 1$ (proofs in Appendix B).

For shell 3 the witness is exactly the house spirit from above, and $m_1 = 3$: from the third shell onwards depth -1 is possible, from the fifteenth -2, from the eighty-fourth -3. The *two-sided fixed points*, finally, are the elements that tidying up can no longer touch – fixed by K_V *and* K_S at once, that is, $n = \mu(x) = \sigma(P)$. Both notions, jump positions and fixed points alike, are counted in Chapter 6.

One fine-grained detail of this non-commutativity deserves mention: the transformation monoid generated by K_V and K_S – the *compactification monoid* – is finite: exactly six elements, all of which are idempotent except $K_V K_S$ (Register S21; proof in Appendix B). The witness with the smallest *value* is again the house spirit $(6,6)$; in the smallest *world* the idempotence of $K_V K_S$ already fails at $(5,25)$.

5.6 Side fidelity

Can the two sides be interwoven – say, adding on the value side and multiplying on the structure side? Law-abiding here means, as agreed at the outset: both operations commutative and associative, multiplication distributes, an additive zero exists. The systematic test of all sixteen combinations – value or shape as payload, conservative or compact as selector, for each of the two operations – then gives a clear answer: exactly two *of these sixteen* are law-abiding, the value-side semiring and its structure-side mirror. (The product family from Theorem 5.4 stands outside this grid – its selectors compute on excesses – and is law-abiding throughout, but stays entirely on the value side.) All eight mixed candidates die of distributivity; the remaining six side-pure ones fail at neutrality (compact addition forgets the operand's horizon) or at associativity (conservative multiplication, the zero problem from the beginning of the chapter).

Example 5.9. The paradigmatic witness, digit by digit. (In this example $\oplus$ and $\otimes$ denote the mixed candidate arithmetic, not the operations from Theorem 5.2; *value-carried* means: the operation reckons with the values, *structure-carried*: with the shapes.) Let $a = b = (1, 1)$ and $c = (2, 5)$, whose shape is $X + 2$ (digits 12). Adding value-carried, $b \oplus c = (3, 6)$; in H_3 the six happens to have the shape $X + 2$ as well (digits 012). Multiplying now structure-carried by a (shape 1), the shape $X + 2$ persists and is evaluated compactly at horizon 2: result $(2, 5)$ – *the value has dropped from 6 to 5*, because the structure multiplication knows only the shape. On the right, by contrast: $a \otimes b = (1, 1)$, $a \otimes c = (2, 5)$, added value-carried $(3, 6)$. Left $(2, 5)$, right $(3, 6)$: distributivity is dead, and one sees exactly why – one side honestly keeps computing with the value, the other lost it in the change of shape.

Theorem 5.10 (side fidelity theorem)**.** *Every law-abiding arithmetic of the payload-carried form considered here – the operations carry value, shape, or fraction, the horizon rules are arbitrary – is side-pure: addition and multiplication both operate on the value side or both on the structure side.*

The restriction to payload-carried operations is essential: on a countable set any semiring structure whatsoever can be defined by transport – the theorem speaks about the arithmetics that *respect* value, shape, or fraction, and about these it speaks completely.

On its exact reach: depending on the case, the proof needs even less than full law-abidingness – one mixed case dies of left distributivity alone, the other of two-sided distributivity (which belongs to every semiring, including non-commutative ones): right distributivity mirrors the chain construction and replaces commutativity. Non-commutative mixed forms are thereby excluded as well; all that remains conceivable is a mixed system that is distributive on *one side* only – a curiosity outside any semiring axiomatics.

The proof deserves to have its core idea told, for it has a punchline (in full in Appendix B). At first things looked bad: individual computations that a mixed arithmetic would have to satisfy are in fact solvable – in part via Pell equations, that is, exactly the kind of number theory that likes to produce exceptions. A direct contradiction seemed out of reach. The decisive weapon is astonishingly plain: the *constants*. If one multiplies a chain of numbers built up by addition with ever larger constant structures, the digit bound drives the evaluation horizons to infinity – and a non-constant polynomial cannot take the same value at arbitrarily many points. That forces the chain links to be constant. Constants, however, die at the next multiplier, because there too the digit bound makes the horizon grow linearly, while distributivity demands linear growth of the value. Both mixed forms break at the very same spot.

By now the theorem even has a third dimension: the fraction identity from Chapter 3 can be interwoven neither with the value side nor with the shape side, and it carries no arithmetic of its own (its addition can exceed one). Side fidelity is a trilogy: three identities, two computational worlds, no mixture – proofs in Appendix B.

The punchline: the source of the rigidity is the digit condition $0 \leq a_i \leq i$ – the very first, most innocent rule of this book. The rule with which everything began is exactly the rule that prevents value and shape from ever merging into one arithmetic. It is not the horizon rules that are rigid; rigid are the identities.

Back to the surface.

A Hori number has three identities: what it is worth, what it looks like, and which fraction it is. This chapter has shown that the first two carry fully fledged, mirror-image computational worlds, that one and the same one works as doorkeeper in both worlds – and, as a proven theorem: that nobody can compute in two worlds at once, while the third carries none of its own. One must choose. That is the deepest theorem Horimetrik has to offer: consistency has a price, and the price is an identity.

Chapter 6

Nine Counting Sequences from Horimetrik

Theories leave fingerprints: number sequences that arise when you count their objects. In the course of this project Horimetrik has produced nine such sequences, and not one of them appears in the OEIS, the great encyclopaedia of integer sequences – which is no proof of novelty, but an indication that nobody has counted here before. All nine are being submitted to the OEIS; once the A-numbers have been assigned, we will enter them here.[1]

6.1 The profiles

The jump positions of the depth law. 3, 15, 84, 533, 3883, 31998, 294875, 3006206, ... The shell in which the commutator depth (Theorem 5.8) first reaches $-g$. The queen of our sequences: for it an asymptotic law is *proved* ($m_g/(g+2)! \to 1$), and it has passed three prediction tests – the last one exactly: $m_7 = 294875$ was forecast before it was computed.

The maximal product height of squares. 1, 6, 22, 87, 407, 2473, 19527, ... How large a horizon can become when you multiply two maximally filled shapes of capacity r (register D8). Meanwhile proved: the sequence grows like $r! \cdot e^{2\sqrt{r}+O(\log r)}$ – a factorial times a gently exploding factor (Appendix B); only the fine structure of the logarithm term is still open.

The shell jumpers. 1, 8, 60, 360, 2520, 21168, 211680, 2268000, ... The number of values that a single leading zero lifts into the next factorial shell. This sequence gave us the $n!/2$ debacle (Chapter 7): for three terms it looked like a half-factorial, at the fourth it no longer did. Its complement, the *non-jumpers* $n! - D(n)$ (5, 16, 60, 360, 2520, 19152, ...), counts as a sequence of the collection in its own right – distinguished by a fully proved closed formula, the digit formula from Appendix B.

The two-sided fixed points. 4, 17, 88, 540, 3960, 32760, ... Numbers whose representation is simultaneously stable under both tidying-up operations (Section 5.5); countable as $n \cdot n!$ minus shell jumpers, for among the $n \cdot n!$ value-compact elements of a shell, exactly those without a digit at its cap fail structure-compactness – and those are the jumper sequences.

[1] Placeholder: A??????. The submission drafts, with jargon-free definitions and programs, are in the project folder `oeis/`.

The single-digit factorials. 1, 2, 5, 7, 11, 23, 29, 39, 59, 119, 143, ... The n for which $n!$ is a single digit in its own shell – completely characterized as $n = (i+1)!/d - 1$ with $1 \leq d \leq i$, exactly i of them per position (Appendix B); position $i = 1$ supplies the trivial initial term $n = 1$. The only one of our sequences that already came with a closed description at the moment of its discovery – and the supplier of nine predictions that came true exactly.

The composition explosion. 1, 4, 23, 6541, 477 149 751, 26 594 371 862 819 905, ... How high the capacity can climb when a maximally filled shape of capacity r is *substituted* into another of the same kind (Chapter 4). The youngest and by far the fastest-growing sequence of the collection: substitution multiplies the degrees and makes the coefficients explode. Thanks to the extremal principle it is nevertheless effortless to compute – a single canonical substitution per term. Its asymptotics are open.

The interchange maxima and zero count. 0, 1, 6, 43, 351, 2873, 26443, 270291, ... and 2, 5, 13, 23, 61, 111, 274, 564, ... The two growth operations – first prepend a zero, then relocate value-preservingly, or the other way round – do not commute. The difference is measured by the *interchange defect* Ω: the value after "grow first, then lift" minus the value after "lift first, then grow". This defect is *never negative*: growing first, then lifting never yields less. The smallest instructive example is the number three as 10 in H_2. Grow first: 010 in H_3 is the *four*; lifted value-preservingly it stays the four. Lift first: the three becomes 003; a zero in front is free, it stays the three. Hence $\Omega = 4 - 3 = 1$. This was long a conjecture and is by now a theorem – the proof runs through a multiplication lemma with a sharp regime boundary. The equality cases, too, have meanwhile been characterized: a number is indifferent to the order exactly when both proof steps are sharp – carry-free plus a sharp lemma – and the zero count obeys a semi-closed counting formula derived from this (confirmed exactly for all measured terms).

6.2 Why submit sequences?

For two reasons, a modest one and a strategic one. The modest one: it is a small, genuine service to mathematics – clean definitions, programs, a proved law. The strategic one: the OEIS is the best novelty detector in the world. Whoever computes $3, 15, 84, 533$ in the future, anywhere, in whatever context, will find these entries – and with them this book. Conversely, the entries link back here. Should one of the sequences nevertheless be unmasked one day as a known quantity in disguise, that too is a result: then we finally know precisely which part of Horimetrik is old mathematics in new clothes – and which is not.

Back to the surface.

Nine counting questions, nine rows of answers that apparently nobody has written down before. Perhaps because they are new. Perhaps only because nobody ever asked. The difference between these two possibilities is exactly what a submission settles – and that is why we submit.

Chapter 7

Dead Ends: What Did Not Work, and Why

Research reports like to tell only the hits. This chapter does the opposite: it collects the roads we walked down and walked back up again. Not out of contrition, but because the dead ends of this project have revealed more about Horimetrik than many a theorem – and because at least one of them later turned out to be a detour rather than a wrong turn.

7.1 The $n!/2$ debacle, or: three data points are not a pattern

Dead end. Idea: At $n = 5, 6, 7$ the shell-jumper sequence showed the values $60, 360, 2520$ – exactly $n!/2$. A formula! *Obstacle:* The fourth data point: $21168 \neq 20160$. *Lesson:* Patterns built from a few terms are hypotheses, not results.

This story has a remarkable epilogue: much later, an exact digit formula for the same sequence fell out of the sky – and it explains in hindsight *why* the false formula held for three terms. For all three hits the digit formula collapses to a single summand, and that summand equals exactly $n!/2$: for 120 and 5040 because they have single-digit representations in their own shells, for 720 because the first maximal digit cuts the formula off right after its first contributing summand. The coincidence was not noise but a genuine structural feature – only one of the digits of $n!$, not of the sequence itself. This is how dead ends sometimes rehabilitate themselves: yesterday's mistake became the footnote of a theorem.

7.2 The small-world trap, six times over

The same trap caught us six times in the course of this project, in ever new disguises: a bound on the compactification commutator looked like ± 1 up to the seventh world (in truth it is unbounded on both sides); its successor, a one-sided bound, withstood random sampling up to the twenty-first world and fell only to *constructed* witnesses; an element of the compactification monoid acted idempotent up to the fourth world, and the smallest counterexample lives in the fifth; and the $n!/2$ pattern you already know. Cases five and six happened at the upper side of the depth law within a single evening: first a pretty triangular-number hypothesis died on its own candidate sweep, before it was even written down – then the threshold that this sweep delivered turned out, in its turn, to be an artefact of too shallow a search. The common lesson is more precise than the usual "test more":

In the Hori space the interesting counterexamples grow factorially slowly.

Whoever claims bounds must construct, not sample.

Incidentally, the same number haunts several of these stories: the six – target of the smallest shell jump (the five, extended by one zero, becomes the six), the value-smallest separating witness of the monoid, the start of the first interesting shell. It is the house spirit of this book.

7.3 Discarded application dreams

Dead end. Idea: Hori addresses instead of IPv6, Hori heaps instead of binary heaps. *Obstacle:* Within a single horizon the Hori space is structurally rigid – address spaces need local elasticity, and the heap pays for the growing branching factor with expensive child comparisons. *Lesson:* The system's strength is the global factorial ordering, not local flexibility. Applications must fit the structure, not the other way round.

Dead end. Idea: Cardinality arguments – is the Hori space "bigger" than the natural numbers? *Obstacle:* It is countable; any enumeration flattens it. *Lesson:* The added value lies not in cardinality but in structure per value. A chessboard does not become more interesting when you number its squares – and not more boring either.

The third application dream came from outside, was the most ambitious – and died on the very first theorem of this book.

Dead end. Idea: Cryptography from capacity (an external proposal, July 2026). A secret shape is pushed through k leading zeros and evaluated value-compactly; only the bare value is published. The attacker, so the hope went, founders on the difference overflow – the way back via antidifferences blows the capacity – and the side fidelity theorem blocks every hybrid shortcut. *Obstacle:* The way back needs neither antidifferences nor mixed arithmetic. From $y = P(N)$, repeated division with remainder pulls the digits down one by one: $c_0 = y \bmod N$ is *exact* (since $c_0 \leq r < N$), then $y_1 = (y - c_0)/N$ delivers the next round with divisor $N-1$, and so on – a handful of divisions, and the "secret" shape lies in the open. This is no ingenious attack but the bijection from Chapter 2 at a shifted evaluation point: the weights $1, N, N(N-1), \ldots$ again form a mixed-radix system, and the digit bounds stay obediently below the divisors. *Lesson:* The factorial ordering does not scramble, it *sorts* – the very bijectivity that makes the system so clean forbids it cryptographic hardness. Two clarifications for the record: the side fidelity theorem forbids mixed *semirings*, not mixed *algorithms* – an attacker may divide, tabulate, and switch representations without ever mixing an arithmetic. And a one-way function needs a secret; Horimetrik has none, it is deliberately bright everywhere. What remains is the more modest role: the horizon calculus of Chapter 4 works as a *measuring instrument* for capacity and noise growth in other people's constructions – not as a source of hardness of its own.

7.4 The tropical mirage – and its resolution

The most instructive dead end deserves the finale. Early on it was noticed that under addition the horizons *want* to behave like a maximum, and under multiplication like a sum – the behaviour of so-called tropical algebra. The direct attempt to compute that way failed on an overflow problem: the value sum can burst the maximum horizon. Discarded.

Months later – in this project's reckoning of time: hours – the idea returned through the back door. The horizon splits into a mandatory part and a freely chosen surplus, and *on the surplus* the tropical arithmetic works flawlessly: the mandatory part absorbs all overflows, the memory computes with maximum and sum. The idea was never wrong. It merely stood in the wrong coordinate system.

Back to the surface.

If this chapter has one message, it is this: discarding is not defeat, it is bookkeeping. Of our nine documented dead ends (the working register lists them as X1 through X9), one later became a theorem – the tropical one –, one became the footnote of a theorem – the $n!/2$ *debacle –, and the rest became precise warning signs. None of that would have happened had we deleted our failed attempts instead of dating them.*

Chapter 8

Open Questions and a Research Programme

Before the final chapter locates the theory on the map of mathematics, an honest stop along the way: not what has been proved – the preceding chapters have done that – but what is open, and in what order it is worth attempting.

8.1 The most immediate questions

The digit patterns of the factorials. The digit formula for the shell jumpers has shifted the question: no longer "how many jumpers are there?", but "how does $n!$ write itself in its own shell?". First patterns are in: the number of leading zeros grows like the inverse function of the factorial, and the single-digit cases are by now *completely* characterized: they are exactly the $n = (i+1)!/d - 1$ with $1 \leq d \leq i$, exactly i of them per position, and the predictions $n = 29, 39, 59, 119$ as well as $143, 179, 239, 359, 719$ all came true (Appendix B). A complete theory of the remaining digit patterns (distribution of the number of nonzero digits, position of the first maximal digit) is still missing.

The fine structure of interchange equality. The positivity of the interchange defect and the characterization of the equality cases are proved (Appendix B); the counting sequence 2, 5, 13, 23, 61, 111, 274, 564, ... obeys a semi-closed formula over the sharp quotients q' from the positivity proof, and its complete digit unfolding has meanwhile been proved: per level two inequalities, a growing excess, one descent (Appendix B). The asymptotics of the zero-count proportion have meanwhile been settled from above: indifference to the order of growth is asymptotically extremely rare – the proportion falls at least like $e^{-(1+o(1))\,n \ln n}$, because sharpness chokes off the digits level by level. What remains open is the matching lower bound.

The upper side of the depth law. The lower side is by now completely proved – jump positions, asymptotics and, as the last piece during the final revision, the exact equality: the maximal depth in shell m is *exactly* the shell gap, and an always serviceable witness is the terminal representation of the value of the maximal polynomial (for shell 3: the house-spirit number $(6, 6)$). All that remains open is the upper side: how fast does the *maximum* of the commutator grow? Here caution is called for, and methodically well-founded caution at that: only the first jump is exhaustively secured (+2 from the ninth world onwards). All later thresholds are merely bounded from above (+3 from the fourteenth at the latest, +4 from the twentieth at the latest, +5 from the twenty-sixth at the latest) – every deepening of the witness search has so far pushed them forwards, and two premature formula hypotheses died at exactly this front, one of them before it was even written down.

8.2 The structural questions

Coupled arithmetics. The side-fidelity theorem forbids mixing the identities; *within* one side, however, the landscape is not classified. The product constructions via the excess supply a whole family; the conservative-compact semiring is demonstrably not one of them. Are there further coupled structures? Is there a complete classification per side?

Addendum: the non-commutative mixed case is closed. This gap, too, fell during the final revision: if one demands, as in every semiring, *both* distributive laws, then right distributivity mirrors the chain construction and replaces commutativity entirely (proof in Appendix B). What remains is only the curiosity of a one-sidedly distributive mixed system – outside any semiring axiomatics, and honestly more a footnote than a question.

Addendum: three-sided side fidelity is proved. During the final revision the last case fell as well (shape addition with fraction multiplication): the chain equation forces the polynomial identity $2A(Y) = Y^{\underline{k}}$, which demands the coefficient $\frac{1}{2}$ – but digits are integers. Together with the two previously settled cases this gives: there are exactly two worlds of arithmetic (value and shape), and the fraction carries no third and mixes into neither. Details in Appendix B. So what remains open here is – nothing at all; the question moves into Chapter 5 as a proved theorem.

Formalization. The side-fidelity theorem is elementary enough for a complete formalization in Lean; the falling factorials are already available in Mathlib. That would be the gold standard of assurance – and a contribution to the formal-methods community into the bargain.

8.3 What the theory wants to be measured against

Finally, the question behind all questions: when would Horimetrik have *failed*, when would it have *succeeded*? Failed, if an existing formalism is found in which horizon, duality and side fidelity are mere renamings – the literature search has not found one, but searches prove nothing. Succeeded, if one of three things happens: if the counting sequences of this book turn up independently elsewhere; if the side-fidelity theorem or the depth law is proved anew in a foreign language; or if someone who has never read this book finds the same delight in the digits of $n!$ in its own shell as we did.

Back to the surface.

A theory is finished when it no longer generates questions. By that standard Horimetrik is delightfully unfinished – and that is exactly how this stop along the way should end: not with a full stop, but with a colon. What remains is the question of the address: where on the map does all of this live?

Chapter 9

Where Horimetrik Lives

A book about a new theory owes its readers an address at the end: where on the map of mathematics does all of this live? The answer only assembled itself completely at the very end of the work, and it is better than we dared to hope. There are, after all, three possible outcomes for a project like this one. An *unknown* ground space would be suspicious – most likely one had simply searched badly. A *known and complete theory* would be the end – one would have repainted old mathematics. The best case is the third: a known ground space carrying new operations and new theorems. That is exactly where we have landed.

9.1 The inhabitants are old acquaintances

Prepend a leading zero to the digits of a Hori number, $e = (0, a_1, \dots, a_n)$, and every position satisfies $0 \le e_j < j$ – and that is, word for word, the definition of an *inversion sequence* of length $n{+}1$. Inversion sequences are in classical bijection with permutations: e_j counts how many earlier entries of a permutation are larger than the j-th. Hence:

Theorem 9.1 (Register S44)**.** *H_n is in bijection with the inversion sequences of length $n{+}1$ and thus with the permutations of $n{+}1$ elements. Through this lens the structure horizon becomes the permutation statistic*

$$\sigma(P_a) = n - \min_{e_j > 0}\big((j{-}1) - e_j\big),$$

and its distribution is known exactly: on each level $1 \le r \le n$ there are precisely $r \cdot r!$ elements (the formula holds for $a \ne 0$; zero has $\sigma = 0$).

So the factorial size of the horizons was never a coincidence: the inhabitants of the Hori worlds *are* permutation codes. This takes nothing away from the theory – it says precisely what the theory is: Horimetrik studies how these well-known codes change their *value*, their *shape*, and their *identity* when their space grows. Relocations, projectors, side fidelity, capacity – all of that is new on the old ground space.

The statistic itself has a pretty reading: $(j{-}1) - e_j$ is the distance of a digit from its upper bound. The literature knows the extreme cases as *tight entries* (digits sitting right at the top); the structure horizon refines this into a measure: how close does *any* digit come to its stop? Whether this statistic corresponds, under the bijection, to a classical permutation statistic is open (Register V13) – its distribution $r \cdot r!$ is, at any rate, almost suspiciously clean.

9.2 The order: a lattice that already exists

Comparing digit sequences position by position turns every horizon into a product of finite chains – a *distributive lattice*. In this order the maximally filled shape M_r is an ordinary element, and the capacity class $\mathcal{F}_r = \{P : \sigma(P) \leq r\}$ is simply the *principal ideal* below it: everything that lies under M_r. This is the cleanest abstract description of the structure horizon we know: $\sigma(P)$ is the first step of the chain $M_0 \leq M_1 \leq M_2 \leq \cdots$ (with $M_0 = 0$) whose principal ideal contains P – and the extremal principle of Chapter 4 becomes, in this language, almost a tautology. The same positionwise order on inversion sequences was recently studied as the *middle order* on permutations, between the weak and the Bruhat order. Whether our projectors K_V, K_S are monotone in this lattice, whether the commutator is an order defect – these are fully formulated follow-up questions addressed to an existing field.

9.3 The calculus: old operators, new bookkeeping

The falling factorials are the house basis of the *finite operator calculus*: there Δ plays the role of the derivative, and the umbral algebra has been studying delta operators and their basic sequences for decades. Structure polynomials and the horizon derivative are therefore very well placed. What Horimetrik adds is a question that is not centre stage there: not *how* an operator changes a polynomial, but *how much minimal structure capacity* that costs. The horizon calculus – Γ_+, $\Gamma_\times$, Γ_Δ, Γ_E, $\Gamma_\circ$, all exact and all attained at the same canonical object – is a capacity bookkeeping on top of an established operator framework. Together with the counting-width reading from Chapter 4, this is the candidate for a self-contained research paper entirely without the Hori narrative.

9.4 The research programme behind it

Three further neighbourhoods we have sighted but not entered; let them be recorded as a programme. *First:* replacing the digit bounds $0 \leq a_i \leq i$ by arbitrary capacity profiles $0 \leq a_i < s_i$ yields the s-inversion sequences studied in combinatorics, together with their geometry (lecture hall polytopes). An *s-Horimetrik* would be a second theory – which profiles admit a polynomial basis, a value–shape duality, a side fidelity, is completely open. *Second:* rook theory writes its polynomials in exactly our basis; which shapes are rook polynomials of boards, and whether M_r possesses a natural board, would be a combinatorial route to our product formulas. *Third:* the counting semantics of the shapes – coloured ordered injections – is the raw material of combinatorial species; a common language for all the interpretations of this book might be found there.

Back to the surface.

The book's conclusion, in three lines: the inhabitants of Horimetrik are known objects – permutation codes in a known lattice, described in the house basis of the calculus of finite differences. The operations on them – two ways of growing, two ways of tidying up, a capacity bookkeeping, a counting width – are new, and so are the theorems about them. A known ground space with new operations and new theorems: for a small theory that began with the reversal of a digit ordering in the dark, that is probably the best address the map has to offer.

Literature and Context

This bibliography is deliberately annotated. The building blocks of Horimetrik – mixed-radix systems, falling factorials, inversion sequences, the calculus of finite differences – are established mathematics, and this book claims nothing else. Each entry therefore states two things: which building block it contributes, and what is *not* to be found there – that is, the point at which Horimetrik goes beyond its source. Whoever wants to check the placement given in Chapter 9 will find the evidence here.

[1] G. Cantor: *Über die einfachen Zahlensysteme*, Zeitschrift für Mathematik und Physik 14 (1869), 121–128. — The factorial base for fractions; this is exactly the fraction-preserving limit space of the third staircase (Chapter 3), but not the shape-preserving world, nor the interplay between them.

[2] D. E. Knuth: *The Art of Computer Programming*, vol. 2, 3rd ed., Addison–Wesley 1997, Section 4.1. — Mixed-radix systems and the factorial number system (factoradic); there the bases grow towards the significant side, and leading zeros are value-neutral.

[3] D. H. Lehmer: *Teaching combinatorial tricks to a computer*, Proc. Sympos. Appl. Math. 10 (1960), 179–193. — Lehmer codes and inversion tables; they explain the factorial size of the horizons and the digit bound $0 \leq a_i \leq i$, but not the horizon-dependent evaluation.

[4] R. L. Graham, D. E. Knuth, O. Patashnik: *Concrete Mathematics*, 2nd ed., Addison–Wesley 1994, Ch. 2. — Falling factorials and the calculus of finite differences; the toolkit of the horizon derivative.

[5] P.-J. Cahen, J.-L. Chabert: *Integer-Valued Polynomials*, AMS Surveys and Monographs 48, 1997. — Polynomials with integer values in the binomial basis; Horimetrik uses only the positive cone and evaluates at the running horizon.

[6] M. Rigo, M. Stipulanti: *Revisiting regular sequences in light of rational base numeration systems*, arXiv:2103.16966. — Abstract numeration systems in which leading zeros are value-sensitive via the radix order: the closest known neighbour of the zero semantics, treated there in formal-language terms rather than algebraically.

[7] G. Kudryavtseva, V. Mazorchuk: *On Kiselman's semigroup*, arXiv:math/0511374; O. Ganyushkin, V. Mazorchuk: *On Kiselman quotients of 0-Hecke monoids*, arXiv:1006.0316. — Monoids of idempotent generators with absorption relations; the neighbourhood of the six-element compactification monoid.

[8] D. Simon: *Mixed radix numeration bases: Horner's rule, Yang-Baxter equation and Furstenberg's conjecture*, arXiv:2405.19798. — Local base changes in mixed-radix systems; a candidate tool for the interchange defect.

[9] S. Corteel, M. A. Martinez, C. D. Savage, M. Weselcouch: *Patterns in Inversion Sequences I*, arXiv:1510.05434. — Inversion sequences ($0 \leq e_j < j$) and their bijection with permutations;

our H_n is, as a set, exactly this space (Chapter 9). What is not there: the reversed valuation, the horizon transitions, and the structure horizon as a statistic.

[10] C. D. Savage, M. J. Schuster: *Ehrhart series of lecture hall polytopes and Eulerian polynomials for inversion sequences*, J. Combin. Theory Ser. A 119 (2012), 850–870. — s-inversion sequences with arbitrary capacity profiles and their geometry; the framework for the s-Horimetrik programme of Chapter 9.

[11] S. Chern, S. Fu, Z. Lin: *Burstein's permutation conjecture, Hong and Li's inversion sequence conjecture, and restricted Eulerian distributions*, arXiv:2209.12137. — Statistics on inversion sequences, among them the *tight entries* ($e_i = s_i - 1$, statistic tig); the structure horizon refines this extreme-case count into a distance measure (Chapter 9).

[12] M. Bouvel, L. Ferrari, B. E. Tenner: *Between weak and Bruhat: the middle order on permutations*, arXiv:2405.08943. — The coordinatewise order on inversion codes as a lattice between the weak and the Bruhat order; the framework in which $\mathcal{F}_r$ becomes a principal ideal and σ a level of ideals.

[13] K. Beauduin: *Operational Umbral Calculus*, arXiv:2407.16348. — Modern finite operator calculus: delta operators and their basic sequences. The framework for Δ and the structure polynomials; what is not there is the capacity bookkeeping Γ_T.

[14] C. Ikenmeyer, I. Pak: *What is in #P and what is not?*, FOCS 2022, arXiv:2204.13149. — Characterizes the *binomial-good* polynomials (nonnegative integer coefficients in the binomial basis) as the *relativizing* polynomial closure operations of #P, and states the binomial basis conjecture (without relativization the same characterization is only conjectural), whose univariate version would already imply P $\neq$ NP. Because of $X^{\underline{d}} = d!\binom{X}{d}$ the shapes of Horimetrik are a proper subclass of the binomial-good polynomials (Chapter 4); what is *not* there is the quantitative layer: the height $\max(d + c_d)$, the $(r+1)!$ hierarchies, and their growth laws.

[15] OEIS Foundation Inc.: *The On-Line Encyclopedia of Integer Sequences*, `https://oeis.org`. — Reference database for the nine counting sequences of this book; A-numbers will be added once assigned.

Appendix A

Reference: Implementation, Data Tables, Register Status

The reference implementation

All enumerations and verifications in this book rest on a small, deliberately unoptimized Python implementation (`horimetrik_reference.py`) with self-tests: encoding and decoding, structure polynomials, both extensions, both compactifications, structure arithmetic. It is complemented by dated experiment scripts whose results are kept in the working register with a status (proved / conjectured / refuted). The project's guiding principle: enumeration produces conjectures and counterexamples, never theorems – and every claimed bound must survive constructed extreme cases, not merely random samples.

The nine counting sequences

Sequence	Initial values	Status
Jump positions m_g ($g{\geq}1$)	$3, 15, 84, 533, 3883, 31998, 294875, \ldots$	Depth law, asymptotics proved
$\beta(r,r)$ ($r{\geq}1$)	$1, 6, 22, 87, 407, 2473, 19527, \ldots$	Asymptotics proved
Jumpers $D(n)$ ($n{\geq}3$)	$1, 8, 60, 360, 2520, 21168, 211680, 2268000$	Digit formula proved
Non-jumpers $N(n)$ ($n{\geq}3$)	$5, 16, 60, 360, 2520, 19152, 151200, \ldots$	Digit formula proved
Fixed points $F(n)$ ($n{\geq}2$)	$4, 17, 88, 540, 3960, 32760$	$= n \cdot n! - D(n)$ (proved)
Single-digit factorials	$1, 2, 5, 7, 11, 23, 29, 39, 59, 119, \ldots$	fully characterized
$\max \Omega$ ($n{\geq}1$)	$0, 1, 6, 43, 351, 2873, 26443, 270291$	$\Omega \geq 0$ proved
$\#\{\Omega = 0\}$ ($n{\geq}1$)	$2, 5, 13, 23, 61, 111, 274, 564$	Equality criterion proved
$\Gamma_\circ(r,r)$ ($r{\geq}1$)	$1, 4, 23, 6541, 477149751, \ldots$	Extremal principle proved

None of these sequences was in the OEIS at the time of writing; the submission is prepared (A-numbers will be added here).

Confirmed predictions

1. Depth -3 of the commutator first near shell 100 (exactly: 84 as a jump position, confirmed in the range up to 200).
2. $m_6 = 31998$ against a predicted ≈ 31930.
3. $m_7 = 294875$ – exactly as predicted.

4. Single-digit factorials at $n = 29, 39, 59, 119$ and $143, 179, 239, 359, 719$ from the (by now complete) characterization $n = (i{+}1)!/d - 1$ – all nine confirmed.

Register status at this version's editorial deadline

44 proved theorems – among them the limit-space theorems of the three staircases, the two dual semirings, the duality theorem and the side-fidelity theorem, the compactification monoid, the depth law, the digit formula, and the interchange positivity –, plus 9 documented dead ends, 3 refuted conjectures of our own, 9 previously unknown counting sequences, and 4 confirmed groups of predictions (12 individual hits, listed above). The current version of the register, conjectures, counterexamples, and changelog accompanies the project's working files.

Appendix B

The Proofs

The chapters of this book tell the ideas behind the proofs; here are the proofs themselves, ordered by chapter. Notation as in the main text; E_n denotes the encoding V_n^{-1}, $\mu(x)$ the minimal horizon, and $\sigma(P) = \max_{c_d>0}(d + c_d)$ (with $\sigma(0) = 0$) the structure horizon.

On Chapter 3: the limit-space theorems

Theorem B.1 (Theorem 3.2, complete). $\varinjlim(H_n, Z_n) \cong \mathcal{S}$ *and* $\varinjlim(H_n, L_n) \cong \mathbb{N}_0$.

Proof. Structure-preserving. For $m \geq n$ let $Z_{n,m}$ be the prepending of $m - n$ zeros; evidently $Z_{m,k} \circ Z_{n,m} = Z_{n,k}$, the system is directed. In the direct limit, $a \in H_n$ and $b \in H_m$ (w.l.o.g. $n \leq m$) count as equal if they have the same image in a common later horizon, i.e. $0^{k-n}a = 0^{k-m}b$, which is equivalent to $b = 0^{m-n}a$. The classes are thus digit sequences modulo leading zeros. The map $[a] \mapsto P_a$ is well defined (a leading zero adds the coefficient 0 at the highest degree), injective (the digit sequence in H_n is the coefficient sequence read from the right and padded to length n; equal polynomials therefore force $b = 0^{m-n}a$) and surjective (every $P \in \mathcal{S}$ is representable from $H_{\sigma(P)}$ onwards).

Value-preserving. Let $L_{n,m} = E_m \circ V_n$. Since $V_m \circ E_m = \mathrm{id}$, we have $L_{m,k} \circ L_{n,m} = E_k \circ V_n = L_{n,k}$. Two elements are identified if and only if $E_k(V_n(a)) = E_k(V_m(b))$ for some k, and since E_k is injective, exactly when $V_n(a) = V_m(b)$: the classes are the fibres of the value map. $[a] \mapsto V_n(a)$ is well defined, injective and surjective (every x occurs as $V_{\mu(x)}(E_{\mu(x)}(x))$). $\square$

On Chapter 3: the third staircase

Theorem B.2 (Cantor coordinate and right extension). *With* $F(a) = \sum_i a_i/(i+1)!$ *we have* $V_n(a) = (n+1)! \cdot F(a)$. *Appending on the right,* $R(a_1, \ldots, a_n) = (a_1, \ldots, a_n, 0)$*, is fraction-preserving, scales the value by exactly* $n+2$*, and* $\varinjlim(H_n, R_n)$ *is the set of finite factorial fractions in* $[0, 1)$.

Proof. Term by term, $a_i\,(n+1)!/(i+1)! = (n+1)! \cdot a_i/(i+1)!$, hence the factorization. Under right appending all digits keep their position and denominator, and the new zero contributes nothing: F is invariant; the value is $(n+2)!\,F = (n+2) \cdot (n+1)!\,F$. In the limit, exactly those digit sequences are identified that differ by appended zeros; each class carries exactly one finite fraction, and every finite factorial fraction occurs. $\square$

On Chapter 4: the horizon derivative

Theorem B.3 (Δ calculus)**.** *The change in value under a structure-preserving extension is $\Delta P(n+1)$ with $\Delta P(X) = P(X{+}1) - P(X)$ and $\Delta\, X^{\underline{d}} = d\, X^{\underline{d-1}}$; a leading zero is value-neutral if and only if P is constant.*

Proof. Under Z, P is preserved and the evaluation point moves from $n{+}1$ to $n{+}2$; the change is by definition $\Delta P(n{+}1)$. For the difference formula: $(X{+}1)^{\underline{d}} = (X{+}1)\, X \cdots (X{-}d{+}2)$ and $X^{\underline{d}} = X \cdots (X{-}d{+}2)\,(X{-}d{+}1)$ share the common factor $X(X{-}1)\cdots(X{-}d{+}2) = X^{\underline{d-1}}$; the remaining factors are $X{+}1$ and $X{-}d{+}1$ respectively, their difference is d, hence $\Delta\, X^{\underline{d}} = d\, X^{\underline{d-1}}$. Value neutrality: $\Delta P = 0$ at all evaluation points in the admissible range forces, since polynomials with infinitely many roots vanish, $\Delta P \equiv 0$, hence P constant; the converse is clear. For the single step: $\Delta P(n{+}1) = 0$ with nonnegative coefficients of ΔP (all basis differences have nonnegative coefficients) and evaluation point in the positive range – since $\deg \Delta P \le n-2$, all $(n{+}1)^{\underline{d}}$ are strictly positive there – likewise forces $\Delta P = 0$. □

On Chapter 4: the horizon calculus

We write $P \preceq Q$ if every coefficient of P in the falling factorial basis is at most the corresponding coefficient of Q, and $M_r(X) = \sum_{d=0}^{r-1} (r{-}d)\, X^{\underline{d}}$ for the maximally filled shape of horizon r.

Theorem B.4 (extremal principle; Register S39)**.** *Let T be a k-ary operator on the structure semiring that is coefficient-monotone in each argument ($P \preceq P'$ implies $T(\ldots,P,\ldots) \preceq T(\ldots,P',\ldots)$). Then*

$$\Gamma_T(r_1,\ldots,r_k) := \max\{\sigma(T(P_1,\ldots,P_k)) \mid \sigma(P_i) \le r_i\} = \sigma\big(T(M_{r_1},\ldots,M_{r_k})\big).$$

Every operator composed from addition, multiplication and Δ is coefficient-monotone.

Proof. $\sigma(P) = \max_{c_d>0}(d + c_d)$ is monotone with respect to $\preceq$, and $\sigma(P) \le r$ is equivalent to the coefficientwise bound $c_d \le r - d$, i.e. to $P \preceq M_r$. Monotonicity of T yields $T(P_1,\ldots,P_k) \preceq T(M_{r_1},\ldots,M_{r_k})$ and hence $\sigma(T(P_1,\ldots,P_k)) \le \sigma(T(M_{r_1},\ldots,M_{r_k}))$; since $\sigma(M_{r_i}) = r_i$, the bound is attained. For the additional claim: addition and Δ are coefficientwise linear with nonnegative constants ($\Delta\, X^{\underline{d}} = d\, X^{\underline{d-1}}$); for multiplication, the linearization coefficients $\binom{p}{j}\binom{q}{j} j!$ of the product formula are nonnegative, and bilinearity transfers the monotonicity. Compositions of monotone operators are monotone. □

Theorem B.5 (difference overflow; Register S40)**.** *For $r \ge 2$, $\Gamma_\Delta(r) = \lfloor (r+1)^2/4 \rfloor - 1$, attained at the single-digit shape $P = (r{-}d)\, X^{\underline{d}}$ with $d = \lfloor (r+1)/2 \rfloor$.*

Proof. By the extremal principle, $\Gamma_\Delta(r) = \sigma(\Delta M_r)$. ΔM_r has the coefficients $b_k = (k{+}1)(r{-}k{-}1)$ for $k = 0,\ldots,r{-}2$, hence

$$\sigma(\Delta M_r) = \max_{0\le k\le r-2} \big[k + (k{+}1)(r{-}k{-}1)\big] = \max_{0\le k\le r-2} (k{+}1)(r{-}k) \;-\; 1.$$

With $u = k{+}1 \in \{1,\ldots,r{-}1\}$, $u(r{+}1{-}u)$ is the product of two numbers with fixed sum $r{+}1$, maximal at $u = \lfloor (r{+}1)/2 \rfloor$ with value $\lfloor (r{+}1)^2/4 \rfloor$; the range is nonempty for $r \ge 2$. The single-digit shape $P = (r{-}d)\, X^{\underline{d}}$ has $\sigma(P) = r$ and $\sigma(\Delta P) = d - 1 + d(r{-}d) = d(r{+}1{-}d) - 1$, maximal at the same d with the same value. (Confirmed exhaustively for all $(r{+}1)!$ shapes up to $r = 8$: values $1, 3, 5, 8, 11, 15, 19$.) □

Theorem B.6 (composition closure; Register S41)**.** *For P, Q with nonnegative integer coefficients in the falling factorial basis, $P \circ Q$ also has nonnegative integer coefficients. Composition is coefficient-monotone in both arguments; by the extremal principle, $\Gamma_\circ(r,s) = \sigma(M_r \circ M_s)$.*

Proof. Since $P \circ Q = \sum_d c_d(P)\,(Q)^{\underline{d}}$ with $(Q)^{\underline{d}} = Q(Q-1)\cdots(Q-d+1)$, it suffices to show that $(Q)^{\underline{d}}$ is an admissible shape. Write $Q = \sum_k a_k X^{\underline{k}}$ and choose pairwise disjoint sets T_k with $|T_k| = a_k$ (*templates of arity k*). Let a *Q-structure* on $[x]$ be a pair (t, φ) consisting of a template $t \in T_k$ and an injection $\varphi\colon [k] \hookrightarrow [x]$; their number is $\sum_k a_k\, x^{\underline{k}} = Q(x)$. Hence $Q(x)^{\underline{d}}$ counts the d-tuples of pairwise distinct Q-structures on $[x]$.

Classify the tuples by the union $U \subseteq [x]$ of the images of all injections involved. The number N_m of tuples whose image union entirely exhausts a fixed m-set depends only on m, and for all $x \in \mathbb{N}_0$ the polynomial identity $Q(x)^{\underline{d}} = \sum_m N_m \binom{x}{m}$ holds. The symmetric group S_m acts on the exhausting tuples by renaming the ground set, $\pi \cdot (t,\varphi) = (t, \pi \circ \varphi)$, and this action is *free*: if π fixes every member of the tuple, then π fixes every image pointwise, hence by exhaustion all of $[m]$, i.e. $\pi = \mathrm{id}$. (Templates of arity 0 do no harm: they contribute nothing to the union.) Thus $m! \mid N_m$, and $(Q)^{\underline{d}} = \sum_m (N_m/m!)\, X^{\underline{m}}$ has coefficients in $\mathbb{N}_0$ – positivity and integrality in one stroke.

Monotonicity in P: an $\mathbb{N}_0$-linear combination of the $(Q)^{\underline{d}}$. Monotonicity in Q: $a_k \le a'_k$ yields $T_k \subseteq T'_k$; every tuple of Q-structures is one of Q'-structures with the same image union, hence $N_m \le N'_m$ term by term. (Closure and extremal principle confirmed computationally and exhaustively for $r, s \le 4$, the closure additionally for 300 random pairs up to $\sigma = 8$.) ☐

Theorem B.7 (shift overflow; Register S42)**.** *For $E\,P(X) = P(X{+}1)$, $\Gamma_E(r) = r + \lfloor r^2/4 \rfloor = \Gamma_\Delta(r{+}1)$.*

Proof. E is linear with nonnegative constants ($(X{+}1)^{\underline{d}} = X^{\underline{d}} + d\,X^{\underline{d-1}}$), so the extremal principle applies: $\Gamma_E(r) = \sigma(E\,M_r)$ with coefficients $c_k = (r{-}k) + (k{+}1)(r{-}k{-}1)$ for $k \le r{-}2$ and $c_{r-1} = 1$; thus $k + c_k = r + (k{+}1)(r{-}k{-}1)$, and the product of two numbers with sum r is at most $\lfloor r^2/4 \rfloor$. The second equality is the identity $r + \lfloor r^2/4 \rfloor = \lfloor (r{+}2)^2/4 \rfloor - 1$. ☐

Theorem B.8 (Hori #P theorem; Register S43)**.** *If $f \in$ #P and P is a shape (nonnegative integer coefficients in the falling factorial basis), then $P \circ f \in$ #P.*

Proof. Let $f(x)$ be the number of witnesses of x under a polynomial verifier V, and $P = \sum_{d=0}^m c_d X^{\underline{d}}$. New machine on input x: guess $d \in \{0, \dots, m\}$, guess a variant $\ell \in \{1, \dots, c_d\}$, guess d witness candidates $w_1, \dots, w_d$; accept if and only if V accepts all w_i and the w_i are pairwise distinct. Since m and all c_d are fixed constants, the machine remains polynomial, and its number of accepting paths is exactly $\sum_d c_d\, f(x)^{\underline{d}} = P(f(x))$, since $f(x)^{\underline{d}}$ counts the ordered d-tuples of pairwise distinct witnesses. (By $X^{\underline{d}} = d!\binom{X}{d}$ the statement also follows from the known characterization of the *binomial-good* polynomials as #P closure operations, Ikenmeyer–Pak, FOCS 2022; the proof above is self-contained.) ☐

On Chapter 5: the two semirings and the duality

Theorem B.9 (Theorem 5.2, complete)**.** *With $(n,x) \oplus (m,y) = (\max(n, m, \mu(x{+}y)), x{+}y)$ and $(n,x) \otimes (m,y) = (\mu(xy), xy)$, $\mathcal{H}$ is a commutative semiring without identity; $(0,0)$ is neutral and absorbing; $h \otimes (1,1) = K_V(h)$; the value-compact elements form a subsemiring isomorphic to $(\mathbb{N}_0, +, \cdot)$ with identity $(1,1)$.*

Proof. Preliminary remark: μ is monotone. *Associativity of* $\oplus$*:* both bracketings of (n,x), (m,y), (k,z) yield the value $x{+}y{+}z$ and the horizons $\max(n,m,k,\mu(x{+}y),\mu(x{+}y{+}z))$ and, respectively, with $\mu(y{+}z)$; since $x{+}y \le x{+}y{+}z$ and $y{+}z \le x{+}y{+}z$, $\mu(x{+}y{+}z)$ absorbs the inner terms. *Associativity of* $\otimes$*:* both sides equal $(\mu(xyz), xyz)$. *Distributivity:* the left-hand side is $(n,x)\otimes\big((m,y)\oplus(k,z)\big) = (\mu(xy{+}xz),\, xy{+}xz)$, the right-hand side $(\max(\mu(xy),\mu(xz),\mu(xy{+}xz)),\, xy{+}xz)$; since $xy \le xy{+}xz$ and $xz \le xy{+}xz$, $\mu(xy{+}xz)$ again absorbs the inner terms. *Neutral/absorbing:* $(n,x)\oplus(0,0) = (\max(n,0,\mu(x)),x) = (n,x)$ since $n \ge \mu(x)$; $(n,x)\otimes(0,0) = (0,0)$. *No identity:* a neutral (k,e) would require $xe = x$ for all x, hence $e = 1$, and $\mu(x) = n$ for all elements – this fails at $(1,0)$. *Projector:* $(n,x)\otimes(1,1) = (\mu(x),x) = K_V(n,x)$. *Corner:* for value-compact operands, $\max(\mu(x),\mu(y),\mu(x{+}y)) = \mu(x{+}y)$; under $\otimes$ the result is value-compact by definition; $(1,1)$ is neutral there, and $x \mapsto (\mu(x),x)$ is the isomorphism. $\square$

Lemma B.10 (σ addition monotonicity)**.** $\sigma(P{+}Q) \ge \max(\sigma P, \sigma Q)$.

Proof. At the maximizing position d^* of P, $c_{d^*}(P{+}Q) \ge c_{d^*}(P) > 0$, hence $\sigma(P{+}Q) \ge d^* + c_{d^*}(P{+}Q) \ge \sigma(P)$. $\square$

Theorem B.11 (Theorem 5.6, complete)**.** *With* $(n,P)\boxplus(m,Q) = (\max(n,m,\sigma(P{+}Q)), P{+}Q)$ *and* $(n,P)\boxtimes(m,Q) = (\sigma(PQ), PQ)$*,* $\mathcal{H}$ *is a commutative semiring without identity;* $(0,0)$ *is neutral and absorbing;* $h \boxtimes (1,1) = K_S(h)$*;* K_S *is a homomorphism for both operations; the structure-compact elements form a subsemiring isomorphic to* $\mathcal{S}$*.*

Proof. Word for word the mirror image of the previous proof, with Lemma B.10 in place of the monotonicity of μ. In addition, the homomorphism property of K_S: for $\boxplus$, $K_S(a)\boxplus K_S(b) = (\max(\sigma P, \sigma Q, \sigma(P{+}Q)), P{+}Q) = (\sigma(P{+}Q), P{+}Q) = K_S(a\boxplus b)$ by the lemma; for $\boxtimes$, both sides equal $(\sigma(PQ), PQ)$. No $\boxtimes$-neutral element: it would require $PE = P$, hence $E = 1$, and $\sigma(P) = n$ for all elements – this fails at $(2,1)$, since $\sigma(1) = 1$. $\square$

Theorem B.12 (product principle, Theorem 5.4)**.** *Every commutative semiring structure* $(\uplus, \odot)$ *on the excesses yields, via* $(x,\varepsilon)\oplus(y,\delta) = (x{+}y, \varepsilon\uplus\delta)$*,* $(x,\varepsilon)\otimes(y,\delta) = (xy, \varepsilon\odot\delta)$*, a law-abiding arithmetic on* $\mathcal{H}$*.*

Proof. The map $(n,x) \mapsto (x,\, n-\mu(x))$ is a bijection $\mathcal{H} \to \mathbb{N}_0\times\mathbb{N}_0$ (the condition $n \ge \mu(x)$ is exactly $\varepsilon \ge 0$; the inverse is $(x,\varepsilon) \mapsto (\mu(x){+}\varepsilon, x)$). The operations are defined componentwise, with $(\mathbb{N}_0, +, \cdot)$ in the value factor; a product of two commutative semirings satisfies all laws componentwise. The selectors are valid, since the resulting horizon is $\mu(\text{value}) + \text{excess} \ge \mu$. $\square$

Theorem B.13 (compactification monoid)**.** K_S *preserves value-compactness. Consequently* $K_V K_S K_V = K_S K_V$*, and the transformation monoid generated by* K_S, K_V *has exactly six elements –* id*,* K_S*,* K_V*,* $K_S K_V$*,* $K_V K_S$ *and* $K_S K_V K_S$ *– of which all except* $K_V K_S$ *are idempotent.*

Proof. *Stability:* let (m,P) be value-compact; the element $(0,0)$ is fixed by K_S, so let $m \ge 1$ and hence $P(m{+}1) \ge m!$. Let $s = \sigma(P)$; suppose we had $P(s{+}1) < s!$. For $s \le t < m$, each term $c_d X^{\underline{d}}$ grows from $t{+}1$ to $t{+}2$ by the factor $(t{+}2)/(t{+}2{-}d)$, hence, since $d \le s{-}1 \le t{-}1$, by at most $(t{+}2)/3$; thus $P(t{+}2) < t!\,(t{+}2)/3 \le (t{+}1)!$, and induction yields $P(m{+}1) < m!$ – contradiction. *Relation:* for every h, $K_V(h)$ is value-compact, by stability so is $K_S(K_V(h))$, hence K_V fixes this element. *Six elements:* every alternating word of length ≥ 4 contains $K_V K_S K_V$ and reduces; squares reduce by idempotence – at most six. Pairwise distinctness by explicit separating witnesses (finite enumeration); in particular $h = (6,6)$ separates: $K_V K_S(h) = (3,6)$, but $K_S K_V K_S(h) = (2,5)$. Hence $(K_V K_S)^2 = K_S K_V K_S \ne K_V K_S$: not idempotent; the idempotence of the remaining five is verified using the relation. $\square$

On Chapter 5: the depth law

Theorem B.14 (Theorem 5.8, complete; Register S33). *For every shell $m \geq 3$,* $\min\{\delta(h) \mid \mu(\mathrm{val}(h)) = m\} = -g(m)$ *with* $g(m) = m - s_{\min}(m)$, $s_{\min}(m) = \min\{s \mid M_s(m{+}1) \geq m!\}$*; the terminal representation of* $x^* = M_{s_{\min}}(m{+}1)$ *is a witness.*

Proof. Bound. Let $h = (n, x)$ with $\mu(x) = m$, so $x \geq m!$. Left: $K_V(h) = (m, x)$, and for $P = \mathrm{struct}(x, m)$ we have $P \preceq M_{\sigma(P)}$ coefficientwise, hence $x = P(m{+}1) \leq M_{\sigma(P)}(m{+}1)$; since $x \geq m!$ it follows that $\sigma(P) \geq s_{\min}$ and thus $\mathrm{hor}(K_S K_V(h)) = \sigma(P) \geq s_{\min}$. Right: for $P_n = \mathrm{struct}(x, n)$ the evaluation is monotone (nonnegative coefficients), hence $\mathrm{val}(K_S(h)) = P_n(\sigma(P_n){+}1) \leq P_n(n{+}1) = x < (m{+}1)!$ and thus $\mathrm{hor}(K_V K_S(h)) \leq m$. Together $\delta(h) \geq s_{\min} - m = -g(m)$.

Attainment. We have $x^* = M_{s_{\min}}(m{+}1) \in [m!, (m{+}1)!)$ – the lower bound by the definition of $s_{\min}$, the upper since $M_{s_{\min}} \preceq M_m$ and $M_m(m{+}1) = (m{+}1)! - 1$. Let $h^* = (x^*, x^*)$ be the terminal representation. K_S fixes h^* (constant shape), hence $\mathrm{hor}(K_V K_S(h^*)) = \mu(x^*) = m$. On the other hand, $M_{s_{\min}}$ is a valid digit sequence in horizon m ($c_d \leq s_{\min} - d \leq m - d$) with value x^*, hence by the bijection from Chapter 2 *the* digit sequence: $\mathrm{struct}(x^*, m) = M_{s_{\min}}$, and $\mathrm{hor}(K_S K_V(h^*)) = \sigma(M_{s_{\min}}) = s_{\min}$. Thus $\delta(h^*) = s_{\min} - m = -g(m)$. □

On Chapter 5: the side-fidelity theorem

Theorem B.15 (Theorem 5.10, value-shape cases). *(A) If $\oplus$ is value-carried and $\otimes$ structure-carried, then already left distributivity is unsatisfiable for every choice of horizon rules. (B) If $\oplus$ is structure-carried and $\otimes$ value-carried, then left distributivity and commutativity of $\otimes$ are jointly unsatisfiable.*

Proof. (A). Let $u = (1, 1)$ (structure 1), $e_1 = u$, $e_m = e_{m-1} \oplus u$; since $\oplus$ is value-carried, we have $\mathrm{val}(e_m) = m$, and the structure Q_m of e_m is some valid representation of m. Left distributivity yields inductively

$$\mathrm{val}(a \otimes e_m) = m \cdot \mathrm{val}(a \otimes u). \tag{B.1}$$

Step 1: Q_m is constant. For $c \geq 1$, $a_c = (c, c)$ has the constant structure c (terminal representation); constants evaluate to themselves everywhere, so $\mathrm{val}(a_c \otimes u) = c$. With (B.1) it follows that $c\, Q_m(y_c{+}1) = mc$, hence $Q_m(y_c{+}1) = m$, where y_c, the horizon of $a_c \otimes e_m$, grows without bound because $y_c \geq \sigma(cQ_m) = \max_d(d + c\, q_d) \geq c$ ($Q_m \neq 0$, since $\mathrm{val}(e_m) = m \geq 1$). All evaluation points lie in the strictly monotone range (arguments $\geq \sigma + 1 \geq \deg{+}2$; falling factorials are strictly increasing there). If $\deg Q_m \geq 1$, Q_m would take the value m at most once – contradiction. Hence Q_m is constant, and with $a = u$ it follows that $Q_m = m$. *Step 2: constants die.* Let $g = (2, 3)$ (structure X), let t_g be the horizon of $g \otimes u$ and $r = \mathrm{val}(g \otimes u) = t_g{+}1 \geq \sigma(X) + 1 = 3$ fixed. By Step 1, $g \otimes e_m$ has the structure $X \cdot m = mX$ with $\sigma(mX) = m{+}1$; on its horizon $T_m \geq m{+}1$ we therefore have $\mathrm{val}(g \otimes e_m) = m(T_m{+}1) \geq m(m{+}2)$; by (B.1) this value is mr – contradiction for $m > r - 2$.

(B). Now structures add and values multiply. Let $f_1 = u$, $f_m = f_{m-1} \oplus u$; then f_m has the constant structure m and the value m. Let A be the structure of $g \otimes u$. Inductively, $g \otimes f_m$ has the structure mA; as a $\otimes$-result of value $3m$, realized on its horizon T_m, the validity of the representation (digit bound) gives $\sigma(mA) \leq T_m$, hence $T_m \geq m$, and $A(T_m{+}1) = 3$ at unboundedly many points. As in Step 1, A is constant, so $A = 3$. Consider $f_2' = g \oplus g$ with structure $2X$, horizon $H \geq \sigma(2X) = 3$, value $2(H{+}1) \geq 8$. Left distributivity and commutativity give $u \otimes (g \oplus g) = (g \otimes u) \oplus (g \otimes u)$ with structure $2 \cdot 3 = 6$ (a constant), hence value 6; but the left-hand side is $1 \cdot 2(H{+}1) \geq 8$. Contradiction. □

Theorem B.16 (noncommutative case)**.** *If $\oplus$ is structure-carried and $\otimes$ value-carried and both distributive laws hold, then the system is unsatisfiable – without commutativity, associativity, or neutral elements.*

Proof. Right distributivity yields inductively $P_{f_m \otimes a} = m \cdot P_{u\otimes a}$; since $f_m \otimes a$ has the value $m\cdot\mathrm{val}(a)$ and the horizon $\geq \sigma(m\, P_{u\otimes a}) \geq m$ grows, $P_{u\otimes a}$ is constant equal to $\mathrm{val}(a)$ (monotonicity argument as above). In particular $P_{u\otimes g} = 3$; the $g \oplus g$ finale then runs word for word as in the commutative case via left distributivity. $\square$

On Chapters 5 and 8: the three-sided side fidelity

Theorem B.17 (trilogy)**.** *Fraction-carried addition is never total; value-carried addition with fraction-carried multiplication, as well as structure-carried addition with fraction-carried multiplication, are unsatisfiable for every choice of selectors (left distributivity suffices).*

Proof. Totality: $F(u) + F(u) = \frac{1}{2} + \frac{1}{2} = 1 \notin [0, 1)$, and every element has fraction < 1.

Value $\oplus$, fraction $\otimes$: chains e_m with $\mathrm{val}(e_m) = m$, horizon $N_m \geq \mu(m)$, fraction $m/(N_m{+}1)!$. Distributivity gives $\mathrm{val}(u \otimes e_m) = m\, w$ with $w = \mathrm{val}(u \otimes u)$ – an integer and ≥ 1, since the fraction of $u \otimes u$ is $\frac{1}{4} > 0$. On the other hand, with h_m the horizon of $u \otimes e_m$, $\mathrm{val}(u \otimes e_m) = \frac{1}{2} \cdot \frac{m}{(N_m+1)!}(h_m{+}1)!$, hence $(h_m{+}1)!/(N_m{+}1)! = 2w > 1$ constant – but a factorial quotient > 1 is at least $N_m{+}2 \to \infty$.

Structure $\oplus$, fraction $\otimes$: chains f_m with constant structure m, horizon $N_m \geq m$, fraction $m/(N_m{+}1)!$. Distributivity gives $P_{u\otimes f_m} = m\, A$ with $A = P_{u\otimes u}$, $\deg A =: d$; comparing fractions on the horizon T_m of $u \otimes f_m$ yields, after cancelling m,

$$2\, A(T_m{+}1) = (T_m{+}1)^{\underline{k_m}}, \qquad k_m = T_m - N_m \geq 1.$$

The product on the right is at least $(N_m + k_m/2)^{k_m/2}$, the left-hand side at most $C(N_m{+}k_m{+}1)^d$; since $N_m \to \infty$, $k_m \leq 2d{+}1$ for large m, some value k^* occurs infinitely often, and $2A(Y) = Y^{\underline{k^*}}$ holds as a polynomial identity. Then A would have the coefficient $\frac{1}{2}$ – digits are integers. (General multipliers a: the same computation with $c = (n_a{+}1)!/v_a > 1$ instead of 2.) $\square$

On Chapter 6: asymptotics and digit formula

Theorem B.18 (asymptotics of the jump positions)**.** *Let m_g be the smallest shell with $M_{m-g}(m{+}1) \geq m!$, where $M_s = \sum_{d<s}(s{-}d)X^{\underline{d}}$ is the maximal polynomial. With $c_g = (g{+}2)! \sum_{k\geq 1} k/(g{+}k{+}1)!$ we have $(g{+}2)!/c_g \leq m_g + 1 \leq (g{+}2)!$ and $c_g = 1 + O(1/g)$, hence $m_g/(g{+}2)! \to 1$.*

Proof. Reindexing ($k = s{-}d$, $s = m{-}g$) yields exactly

$$\frac{M_{m-g}(m{+}1)}{m!} = (m{+}1) \sum_{k=1}^{m-g} \frac{k}{(g{+}k{+}1)!} =: F(m).$$

F grows in m (the prefactor strictly, the sum through newly arriving nonnegative terms), so the threshold is well defined. For $m{+}1 \geq (g{+}2)!$ the term $k{=}1$ suffices: $F(m) \geq (m{+}1)/(g{+}2)! \geq 1$. Conversely, $F(m) \leq (m{+}1)\, c_g/(g{+}2)!$, hence $F(m) < 1$ for $m{+}1 < (g{+}2)!/c_g$. Finally, with $x = 1/(g{+}3)$,

$$c_g \leq 1 + \sum_{k\geq 2} k\, x^{k-1} = 1 + \frac{x(2-x)}{(1-x)^2} = 1 + O(1/g). \qquad \square$$

Theorem B.19 (digit formula for the non-jumpers)**.** *Let $d = (d_1, \dots, d_n)$ be the H_n digit sequence of $n!$ and P the first position with maximal digit ($d_P = P$), or $P = n$ if none exists. Then*

$$N(n) = \sum_{p=1}^{P} \min(d_p, p) \cdot \frac{n!}{p!}.$$

Proof. $N(n)$ counts digit sequences with $a_i \le i-1$ and value $< n!$. We decompose by the first position p at which a differs from d. The prefix must agree with d – possible only as long as $d_j \le j-1$, i.e. up to and including the first maximal digit. At position p we need $a_p < d_p$ and $a_p \le p-1$: $\min(d_p, p)$ possibilities. The suffix is free within the caps: $\prod_{j>p} j = n!/p!$ possibilities, and every such suffix stays strictly below, because the full (uncapped) suffix space fills the interval $[0, w_p)$ with the place weight $w_p = (n{+}1)!/(p{+}1)!$ bijectively. Every non-jumper is counted exactly once. ☐

Theorem B.20 (β reduction)**.** $\ln \beta(r,r) = \ln \max_t \binom{r-1}{t}^2 (r{-}1{-}t)! + O(\log r)$.

Proof. Lower bound: the triple $(p,q,j) = (r{-}1, r{-}1, r{-}1{-}t)$ contributes to degree $r{-}1{+}t$ the summand $\binom{r-1}{t}^2 (r{-}1{-}t)!$, and all contributions are nonnegative. Upper bound: each contribution is at most $r^2 \binom{r-1}{j}^2 j!$, and per degree there are at most r^3 triples; hence $\beta \le 2r + r^5 \max_t(\cdot)$. ☐

Theorem B.21 (β asymptotics)**.** $\ln(\beta(r,r)/r!) = 2\sqrt{r} + O(\log r)$.

Proof. By the reduction theorem it suffices to prove the statement for the maximal term $\binom{s}{t}^2 (s-t)!/r! = s!/(r\, t!^2 (s-t)!)$ with $s = r-1$. From above: Stirling bounds give $\ln(s!/(t!^2(s-t)!)) \le -h(t) + O(\log s)$ with $h(t) = 2t\ln t - t + (s{-}t)\ln(s{-}t) - s\ln s$; concavity yields $h(t) \ge t(2\ln t - \ln s - 2)$, and this function has its minimum $-2\sqrt{s}$ at $t = \sqrt{s}$. From below: $t = \lceil \sqrt{s} \rceil$ with $\binom{s}{t} \ge (s-t)^t/t!$ gives $t\ln((s-t)/t^2) + 2t - O(\log r) = 2\sqrt{s} - O(\log r)$. ☐

Theorem B.22 (single-digit factorials)**.** *$n!$ is a single digit in H_n if and only if $n = (i{+}1)!/d - 1$ with $1 \le d \le i$; the digit is d at position i, and per position there are exactly i cases.*

Proof. A single digit d at position i means $n! = d\,(n+1)!/(i+1)!$, hence $d = (i+1)!/(n+1) \le i$. Conversely, every $d \le i$ divides $(i+1)!$, and $n+1 = (i+1)!/d$ makes the equation exact; the position exists, since $n{+}1 = (i{+}1)!/d \ge (i{+}1)!/i \ge i{+}1$ implies $i \le n$, and the unique digit expansion then consists of exactly this digit. d runs bijectively through $1, \dots, i$. ☐

On Chapters 6 and 8: interchange positivity, sharpness, zero count

For $x < (n{+}1)!$ let $z_n(x) := P(n{+}2)$ with $P = \mathrm{struct}(x, n)$ – the value after one zero extension. The *interchange defect* of an element $h = (n, x)$ is $\Omega(h) = z_n(x) - z_{n+1}(x)$: the value after "grow first, then lift value-preservingly" minus the value after "lift first, then grow".

Lemma B.23 (recursion formula)**.** *For $x = q(n{+}1) + r$, $0 \le r \le n$: $z_n(x) = r + (n{+}2)\, z_{n-1}(q)$.*

Proof. From the digit recursion it follows that $P_{n,x}(X) = r + X \cdot P_{n-1,q}(X{-}1)$, since the coefficient shift by one degree is, by $X^{\underline{d}} = X \cdot (X{-}1)^{\underline{d-1}}$, multiplication by X with a shift of argument. Evaluate at $X = n{+}2$. ☐

Lemma B.24 (strict monotonicity and superadditivity)**.** *$z_n(x) < z_n(x{+}1)$, consequently $z_n(y{+}s) \ge z_n(y) + s$.*

Proof. Induction on n; base $n = 1$: $z_1(0) = 0 < 1 = z_1(1)$. Without a carry, z_n grows by exactly 1; with a carry ($r = n$), $z_n(x{+}1) - z_n(x) = (n{+}2)(z_{n-1}(q{+}1) - z_{n-1}(q)) - n \geq (n{+}2) - n = 2$. $\square$

Lemma B.25 (multiplication lemma)**.** *For $c \geq \ell{+}2$ and $cy < (\ell{+}1)!$, $z_\ell(cy) \geq (c{+}1)\, z_\ell(y)$.*

Proof. Induction on ℓ; the base cases are empty and trivial, respectively. Let $y = q(\ell{+}1) + r$ and $cr = s(\ell{+}1) + u$ with $0 \leq u \leq \ell$; then $cy = (cq{+}s)(\ell{+}1) + u$ and, by the recursion formula, $z_\ell(cy) = u + (\ell{+}2)\, z_{\ell-1}(cq{+}s)$. Superadditivity and the induction hypothesis (the regime is preserved: $c \geq \ell{+}2 > (\ell{-}1){+}2$; validity: $cq < \ell!$) give $z_{\ell-1}(cq{+}s) \geq (c{+}1)z_{\ell-1}(q) + s$. It remains to show $u + (\ell{+}2)s \geq (c{+}1)r$: with $c = \ell{+}1{+}e$, $e \geq 1$, we have $u = er \bmod (\ell{+}1)$, and the inequality is equivalent to $r(c{-}\ell{-}1) = er \geq er \bmod (\ell{+}1) = u$ – true. $\square$

Theorem B.26 (interchange positivity)**.** $\Omega(h) \geq 0$ *for all $h \in \mathcal{H}$.*

Proof. By the definition of Ω it must be shown that $z_{n+1}(x) \leq z_n(x)$ for $x < (n{+}1)!$. Let $x = q'(n{+}2) + r'$. Then

$$z_{n+1}(x) = r' + (n{+}3)\, z_n(q') \leq r' + z_n\big((n{+}2)q'\big) \leq z_n\big(q'(n{+}2) + r'\big) = z_n(x),$$

first by the multiplication lemma ($c = n{+}2$, exactly at the regime boundary), then by superadditivity. $\square$

Theorem B.27 (equality cases; Register S32)**.** $\Omega(h) = 0$ *holds if and only if both steps of this chain are sharp: the multiplication lemma with $c = n{+}2$ for the sharp quotient q', i.e. $z_n((n{+}2)q') = (n{+}3)\, z_n(q')$, and superadditivity carry-free, i.e. $z_n((n{+}2)q'{+}r') = z_n((n{+}2)q'){+}r'$.*

Proof. $\Omega(h)$ is the sum of the two deviations $z_n((n{+}2)q') - (n{+}3)z_n(q')$ and $z_n((n{+}2)q' + r') - z_n((n{+}2)q') - r'$; both are nonnegative by the multiplication lemma and superadditivity respectively, so their sum is zero if and only if both are zero. $\square$

Theorem B.28 (digit unfolding of sharpness)**.** *For $c \geq \ell + 2$, $cy < (\ell + 1)!$, $y = q(\ell + 1) + r$, $e = c - \ell - 1$ we have: $z_\ell(cy) = (c{+}1)z_\ell(y)$ if and only if $er \leq \ell$, $(cq \bmod \ell) + r \leq \ell - 1$, and the same condition holds recursively for $(\ell{-}1, c, q)$ (termination: for $\ell = 0$ the condition is empty).*

Proof. The chain $z_\ell(cy) = u + (\ell + 2)z_{\ell-1}(cq + s)$ from the proof of the multiplication lemma is sharp throughout exactly then: the arithmetic estimate when $er \leq \ell$ (then $s = r$, $u = er$), superadditivity when the $s = r$ single steps starting from cq are carry-free – their units digit is $cq \bmod \ell$, the condition therefore $(cq \bmod \ell) + r \leq \ell - 1$ –, the third recursively by definition. Verified exhaustively for $\ell \leq 7$. $\square$

Theorem B.29 (zero-count bound)**.** $\#\{\Omega = 0$ *on* $H_n\} \leq (n{+}1)^{n+1}/n!$, *so the proportion falls at least like $e^{-(1+o(1))n \ln n}$.*

Proof. By the equality theorem, every such element is determined by (q', r') with sharp q', and carry-freeness forces $r' \leq n$ (the units digit grows by one per step and must stay below its maximum n before each step). By the digit unfolding, sharpness forces at level ℓ the digit bound $r_\ell \leq \lfloor \ell/(n{+}1{-}\ell) \rfloor$, and $\lfloor \ell/j \rfloor + 1 \leq (n{+}1)/j$ with $j = n{+}1{-}\ell$ yields the product $(n{+}1)^n/n!$ for the q'; together with the $n{+}1$ possibilities for r' the bound follows. $\square$

On Chapter 9: the inversion-sequence classification

Theorem B.30 (Register S44)**.** *H_n is in bijection with the inversion sequences of length $n{+}1$ and with S_{n+1}. The structure horizon is $\sigma(P_a) = n - \min_{e_j>0}((j{-}1) - e_j)$ for $a \neq 0$ (for $a = 0$, $\sigma = 0$) with $e = (0, a_1, \dots, a_n)$, and $|\{a \in H_n : \sigma(P_a) = r\}| = r \cdot r!$ for $1 \leq r \leq n$.*

Proof. Setting $e_1 = 0$ and $e_{i+1} = a_i$, we get $0 \leq e_j < j$ for all $j = 1, \dots, n{+}1$ – the defining condition of an inversion sequence; the map $\pi \mapsto (e_j)_j$ with $e_j = |\{i < j : \pi_i > \pi_j\}|$ is the classical bijection to S_{n+1}, and both sides have $(n{+}1)!$ elements. For the structure horizon: $c_d = a_{n-d} = e_{n-d+1}$, hence $d + c_d = (n{+}1 - j) + e_j$ with $j = n - d + 1$; maximizing $d + c_d$ over $c_d > 0$ is minimizing $(j{-}1) - e_j$ over $e_j > 0$, and $\max(d + c_d) = n - \min_{e_j>0}((j{-}1) - e_j)$. Counting: $\sigma(P_a) \leq r$ means $a_i \leq r - (n - i)$ for all i with $a_i > 0$, equivalently $a \leq M$ coordinatewise for the maximal profile belonging to H_r; the number of such a in H_n is $(r{+}1)!$ (the digits above position r are forced to be zero, below they are free up to the bound). Hence $|\{\sigma \leq r\}| = (r{+}1)!$ and $|\{\sigma = r\}| = (r{+}1)! - r! = r \cdot r!$. (Confirmed exhaustively up to $n = 6$, `experiments_einordnung.py`.) □